Communications

Classroom Workbook

ELECTRONICS AND COMM. ENGINEERING

GATE/ESE

unacademy

Communications

Classroom Workbook

For ECE

Copyrights © All Rights Reserved

Sorting HAT Technologies Pvt. Ltd.

No part of this publication may be reproduced or distributed in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise or stored in a database or retrieval system without the prior written permission of the publishers. The program listings (if any) may be entered, stored and executed in a computer system, but they may not be reproduced for publication.

Printing of books passes through many stages - writing, composing, proof reading, printing etc. We try our level best to make the book error- free. If any mistake has inadvertently crept in, we regret it and would be deeply indebted to those who point it out. We do not take any legal responsibility.

Edition : COMM-2024/2025

Sorting HAT Technologies Pvt. Ltd.

353, 12th 'A'cross, off double road, Opp. City Centre

Residency, Indira nagar, 1st stage Bangalore

Bangalore KA560038 IN.

GATE Syllabus

ECE

Random processes: auto correlation and power spectral density, properties of white noise, filtering of random signals through LTI systems. Analog communications: amplitude modulation and demodulation, angle modulation and demodulation, spectra of AM and FM, super heterodyne receivers. Information theory: entropy, mutual information and channel capacity theorem. Digital communications: PCM, DPCM, digital modulation schemes (ASK, PSK, FSK, QAM), bandwidth, inter-symbol interference, MAP, ML detection, matched filter receiver, SNR and BER. Fundamentals of error correction, Hamming codes, CRC.

ESE Syllabus

E & T

Random signals, noise, probability theory, and information theory; Analog versus digital communication and applications: systems (AM, FM, transmitters and receivers), theory, practice and standards; SNR comparison; Digital communication basics: sampling, quantizing, coding, PCM, DPCM, multiplexing audio/video; Digital modulation: ASK, FSK, PSK; Multiple access: TDMA, FDMA, CDMA; Optical communication: fiber optics, theory, practice and standards.

Contents

Chapters

1. Amplitude Modulation
2. Angle Modulation
3. Receiver
4. Pulse Modulation Schemes
5. Digital Shift Keying Techniques
6. Random Variables and Noise
7. Information Theory and Coding

1 Amplitude Modulation

MCQ, MSQ and NAT

Q.1 Consider a modulating signal m(t) = cos ($2\pi f_m t$), where ($f_m << f_c$).

The possible SSB AM signal(s) is/are-

(A)$A_c \sin 2\pi(f_c - f_m)t$

(B)$A_c \cos 2\pi(f_c + f_m)t$

(C)$A_c \sin 2\pi(f_c + f_m)t$

(D)$A_c \cos 2\pi(f_c - f_m)t$

Q.2 Consider a sinusoidal modulating signal given as-

$m(t) = A_m \cos 2\pi f_m t$, where $f_m << f_c$.

The power in the lower sideband of DSB-SC AM is-

(Consider A_C to be the maximum amplitude of the carrier signal)

(A) $\frac{A_c^2 \mu^2}{16}$

(B) $\frac{A_c^2 \mu}{4}$

(C) $\frac{A_c^2 \mu}{8}$

(D) $\frac{A_c^2 \mu^2}{8}$

Q.3 The AM system used for the broadcasting purposes is-

(A)DSB-SC

(B)Conventional AM

(C)SSB

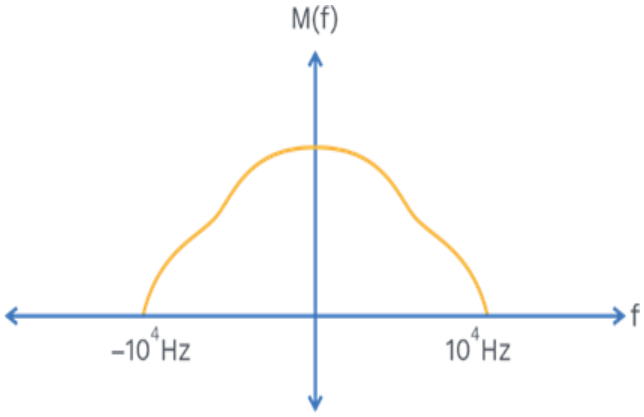
(D)None of these

Q.4 The ratio of power in the carrier to the sidebands, if the output signal from an AM modulator is s(t) = 6 cos (1800 πt) + 24 cos (2000 πt) + 6 cos (2200 πt), is_____ (Rounded off to the nearest integer).

Q.5 An AM signal given by s(t) = [20 + 12cos 8000πt + 10cos 9000πt]cos $2\pi f_c t$. The ratio of the power contained in the sidebands to the total power is_____ (Up to two decimal places).

[Consider the carrier frequency as 100 kHz]

Q.6 Consider a signal m(t) whose Fourier transform is illustrated in the given figure. This signal is transmitted, and it is known that the signal is normalized (–1 ≤ |m(t)| ≤ 1).



The bandwidth (in kHz) of the modulated signal if DSB is employed is_____ (Rounded off to the nearest integer).

Q.7 The correct relation between the bandwidth of the various AM schemes is-

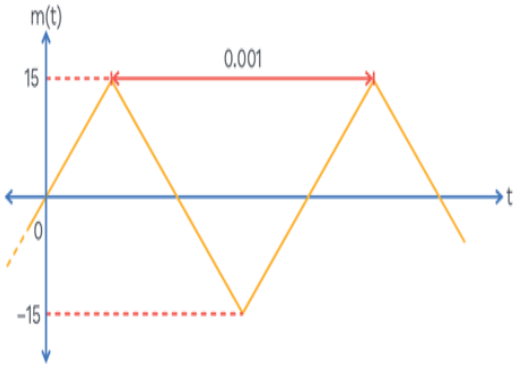
(A)$BW_{(SSB)} < BW_{(VSB)} < BW_{(DSB)}$

(B)$BW_{(DSB)} > BW_{(SSB)} > BW_{(VSB)}$

(C)$[BW_{(VSB)} = BW_{(SSB)}] < BW_{(DSB)}$

(D)$BW_{(SSB)} > BW_{(VSB)} > BW_{(DSB)}$

Q.8 For an AM signal, the message signal with m(t) is expressed as-



If the modulation index is 0.8, then the power efficiency or transmission efficiency (η) is-

(A)19.66%

(B)24.24%

(C)44.96%

(D)33.33%

Q.9 Which of the following statement(s) is/are **correct** for the Hilbert transform?

(A)A signal x(t) and its Hilbert transform $x_h(t)$ have the same autocorrelation function.

(B)Hilbert transform is similar to a $\left(+\frac{\pi}{2}\right)$ phase shift for the positive frequencies and $\left(-\frac{\pi}{2}\right)$ phase shift for the negative frequencies.

(C)Hilbert transform can be expressed in terms of a convolution integral.

(D)The Hilbert transform can be used for generating SSB signals.

Q.10 In which condition(s) the baseband or modulating signal will be preserved in the envelope of the AM signal?

(A)Modulation index, $\mu = 1$

(B)Modulation index, $\mu < 0$

(C)Modulation index, $\mu > 1$

(D)Modulation index, $\mu < 1$

Q.11 Which one of the following statements is correct about the carrier suppression in the DSB signal?

(A)It transfers all the average transmitted power to the sidebands and the DSB scheme makes better use of the average power available from a given transmitter.

(B)By suppressing the carrier and one sideband, the DSB signal power wastage can be saved in DSB-SC.

(C)It reduces the bandwidth of the DSB signal.

(D)None of the above

Q.12 Which of the following is **NOT** a need for modulation in a communication system?

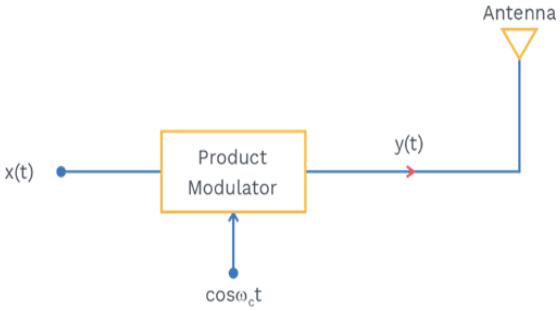
(A)To achieve practical transmitting antenna height.

(B)For transmission of low frequencies.

(C)To allow mixing up of signals from different transmitters.

(D)To reduce noise.

Q.13 Which of the following signal will be obtained at the output of the product modulator?



(A)SSB signal

(B)DSB-SC signal

(C)Conventional AM signal

(D)FM signal

Q.14 The Hilbert transform of $g(t) = \frac{1}{t}$ is -

(A)$\pi\delta(t)$

(B)$\frac{1}{\pi}\delta(t)$

(C)$-\frac{1}{\pi}\delta(t)$

(D)$-\pi\delta(t)$

Q.15 The percentage of the total power carried by the sidebands of the AM wave for single tone modulation, when $\mu = 0.35$ is_____%. (Up to two decimal places)

Q.16 Which of the following is/are the correct reason(s) for the wide use of VSB in broadcast television?

(A)The circuitry for the demodulation must be simple and cheap in the receiver.

(B)Video signal occupies a large bandwidth and low-frequency content.

(C)VSB transmission reduces the overall requirement of the bandwidth.

(D)None of the above.

Q.17 Consider the message signal given below.

$$m(t) = \frac{1}{3}\cos\omega_1 t - \left(\frac{1}{3}\right)\sin\omega_2 t$$

It is used to amplitude modulate the carrier of frequency ω_c to generate

$$s(t) = [1 + m(t)]\cos\omega_c t$$

The power efficiency achieved by this modulation scheme is_____%. [Rounded off to the nearest integer]

Q.18 Five message signals having bandwidths of 4kHz, 8kHz, 10kHz, 10kHz & 5kHz are applied to AM, SSB, DSB, SSB, and AM modulators respectively, and transmitted through a channel using an envelope detector. The guard band is 2 kHz. The bandwidth for transmission of the multiplexed signal is-

(A)56 kHz

(B)64 kHz

(C)72 kHz

(D)128 kHz

Q.19 The output signal from an AM modulator is given as-

s(t) = 4cos4000πt + 10cos4500πt + 4cos5000πt

Find the modulation index and the ratio of carrier power to the total sideband power.

(A)0.6 and 2.232 respectively

(B)0.8 and 3.125 respectively

(C)0.5 and 2.250 respectively

(D)0.75 and 2.922 respectively

Q.20 A signal $x(t) = A\left[\cos 2\pi \times 10^3 t + \cos 8\pi \times 10^3 t\right]$ is

amplitude modulated with carrier signal to generate $x_{AM}(t) = [1 + x(t)] \cos 10^5 t$. If the power efficiency is 20%, then the value A is_____.(Up to 1 decimal place)

Q.21 The Hilbert transform of x(t) is defined as $\hat{x}(t)$. The correct statement(s) is/are

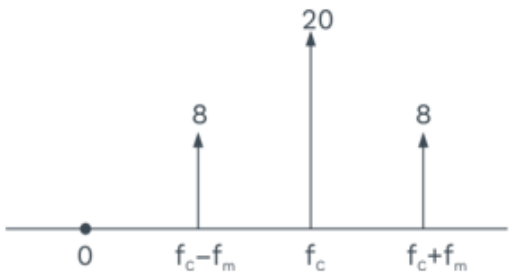
(A)It is a 180° phase shifter.

(B)If a signal x(t) is band-limited, then the Hilbert transform $\hat{x}(t)$ is not band-limited.

(C)The signal x(t) and $\hat{x}(t)$ have the same magnitude spectrum.

(D)The signal x(t) and $\hat{x}(t)$ have the same autocorrelation function.

Q.22 The ratio of total sideband power to the carrier power for the following frequency spectrum of AM is _____.



Q.23 An AM broadcast station transmits 40 kW as average output carrier power. Modulation index used is 0.646 for sine wave modulation. The peak amplitude of the output, if the given antenna is represented by a 40Ω load, is _____kV.

(up to 3 decimal places)

Q.24 The signal $2[1 + 0.4\,\mathrm{Cos}(6000\pi t)]\,\mathrm{Cos}\,(10^6\pi t)$ is applied to a square law device described by $Y = (X + 4)^2$, where Y is the output and X is the input. The output of the square law device is filtered by an ideal low pass filter with a cut-off frequency of 8 KHz. The spectral components at the output of the LPF are

(A)5 KHz & 600 KHz

(B)5 KHz & 3 KHz

(C)3 KHz & 600 KHz

(D)3 KHz & 6 KHz

Q.25 An AM signal is given as $S_{AM}(t) = 4(1 + 0.5 \cos 1000t) \cos 10^6 t$. For proper envelope detection through an envelope detector, R_LC should be

(A)200 ms

(B)8 ms

(C)$600\ \mu s$

(D)$5\ \mu s$

Q.26 Find the modulation index of the following analog modulated signal -

$$S(t) = 10\left[1 + 0.8\cos 4 \times 10^3 \pi t\right] \cos\left(60 \times 10^3 \pi t\right)$$

(A)0.4

(B)0.2

(C)0.6

(D)0.8

Q.27 If a modulating signal with frequency and amplitude 8 kHz and 6 V amplitude modulates a carrier signal with frequency and amplitude 40 kHz and 10V. Find the power of transmitted signal through a resistance of 5Ω.

(A)11.8 W

(B)12 W

(C)10 W

(D)13.6 W

Q.28 If the highest frequency component of an AM signal is 400 kHz, the bandwidth required to transmit this signal is 80 kHz, the carrier frequency is–

(A)240 kHz

(B)120 kHz

(C)360 kHz

(D)480 kHz

Q.29 Find the effective modulation index if the carrier is amplitude modulated by three sinusoidal baseband signals with modulation indices of 0.3, 0.5 and 0.6.

(A)0.875

(B)0.836

(C)0.75

(D)0.735

Q.30 In an amplitude modulated system, the total power is 800W and power in the carrier is 600 W. The modulation index(up to 3 decimal places) is-

(A)0.666

(B)0.333

(C)0.816

(D)0.712

Q.31 The most efficient bandwidth scheme among the given option is-

(A)AM

(B)PM

(C)SSB-SC

(D)FM

Q.32 The power of unmodulated carrier wave if a broadcast radio transmitter radiates 2.5 × 10^3 W and 0.60 modulation index will be -

(A)1.8×10^3 watts

(B)1.18×10^3 watts

(C)4.22×10^3 watts

(D)2.11×10^3 watts

Q.33 In an AM transmission (DSB + Carrier), the modulation index is changed from μ to k/μ. The ratio $\left(\frac{\mathrm{P_{SB}}}{\mathrm{P_C}}\right)$ increases by a factor of 16. The value of k is (P_{SB} and P_C have their usual meanings).

(A)k = 2

(B)k = 6

(C)k = 3

(D)k = 4

Q.34 Find the total power in the amplitude modulated wave, if a 400 mW carrier is modulated to a depth of 75%.

(A)0.6263 W

(B)0.5125 W

(C)0.8423 W

(D)0.528 W

Q.35 A carrier signal $c(t) = 50\cos 2\pi \times 10^5$ is amplitude modulated by an audio frequency signal $x(t) = 10\cos 1000\pi t$. If the modulation index is $\mu = 0.4$, the sideband frequencies are-

(A)100500 Hz, 99500 Hz

(B)500100 Hz, 50099 Hz

(C)20256 Hz, 166750 Hz

(D)25620 Hz, 170166 Hz

Q.36 Let $m(t) = \cos\left[\left(2\pi \times 10^2\right) t\right]$ be the message signal and c(t) be the carrier signal such that $c(t) = 25\cos\left[\left(2\pi \times 10^6\right) t\right]$. The message signal is to be amplitude modulated by using the carrier signal. The modulation index of the generated AM signal is 0.65. The ratio of upper sideband power to the carrier power is—

(A)0.333

(B)0.2

(C)0.105

(D)0.166

Q.37 A carrier signal c(t) = cos 2π f_ct where f_c = 2 MHz is amplitude modulated with a message signal m(t) = cos 2000 πt + 4 cos 4000 πt. For demodulating the generated AM signal using an envelope detector, the time constant RC of the detector circuit should satisfy-

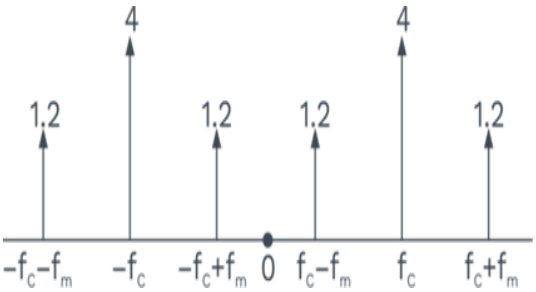
(A)$1\mu s \leq RC < 0.5 \text{ ms}$

(B)$0.5\mu s \leq RC \leq 0.5 \text{ ms}$

(C)$0.5\ \mu s \leq \text{RC} < 1 \text{ ms}$

(D)$\text{RC} \geq 0.5 \text{ ms}$

Q.38 In the given following frequency spectrum [single-tone AM]



If carrier amplitude is doubled find the carrier power (given R = 1Ω)-

(A)32 W

(B)64 W

(C)256 W

(D)128 W

Ans.
D

Answer Key

Q.1 [B, D]

Q.2 [D]

Q.3 [B]

Q.4 [8]

Q.5 [0.23]

Q.6 [20]

Q.7 [A]

Q.8 [B]

Q.9 [A, C, D]

Q.10 [A, D]

Q.11 [A]

Q.12 [C]

Q.13 [B]

Q.14 [D]

Q.15 [5.77]

Q.16 [A, B, C]

Q.17 [10]

Q.18 [B]

Q.19 [B]

Q.20 [0.5]

Q.21 [C, D]

Q.22 [0.32]

Q.23 [2.943]

Q.24 [D]

Q.25 [C]

Q.26 [D]

Q.27 [A]

Q.28 [C]

Q.29 [B]

Q.30 [C]

Q.31 [C]

Q.32 [D]

Q.33 [D]

Q.34 [B]

Q.35 [A]

Q.36 [C]

Q.37 [B]

Q.38 [D]

2 Angle Modulation

MCQ, MSQ and NAT

Q.1 Consider that an FM modulator is given an input signal, $x(t) = 20\cos(10^4\pi t)$ V. If the modulation index is 24, then the sensitivity constant will be _____ kHz/V. (Round off to the nearest integer)

Q.2 A message signal $m(t) = 6\cos(5000\pi t)$ angle modulates a carrier signal $c(t) = A_c \cos(10^6\pi t)$. The deviation constants for PM and FM are 3 rad/V and 3000 Hz/V respectively.

If the maximum amplitude of m(t) is halved, then the bandwidth (in kHz) for PM and FM are respectively-

(A)50 and 23

(B)25 and 50

(C)23 and 50

(D)50 and 25

Q.3 Calculate the peak frequency deviation and modulation index for the FM modulator with a frequency sensitivity of 6 kHz/V. Assume that a modulating signal 10cos (6000 πt) is applied.

(A)Frequency deviation = 20 kHz

Modulation index = 6

(B)Frequency deviation = 20 Hz

Modulation index = 60

(C)Frequency deviation = 60 kHz

Modulation index = 20

(D)Frequency deviation = 60 Hz

Modulation index = 20

Q.4 The frequency deviation in a phase-modulated signal is-

(A)directly proportional to only the amplitude of modulating signal.

(B)directly proportional to only the frequency of modulating signal.

(C)inversely proportional to both amplitude and frequency of modulating signal

(D)directly proportional to both amplitude and frequency of modulating signal.

Q.5 Choose the incorrect statement from the statements given in the options-

(A)The amplitude of the FM wave is constant. It is independent of the modulation index.

(B)FM receivers are more immune to noise than AM receivers.

(C)In FM, the number of sidebands having significant amplitude depends on the modulation index.

(D)In FM, the transmitted power is directly proportional to the modulation index.

Q.6 Choose the correct statement(s) regarding a balanced slope detector.

(A)The characteristics of a balanced slope detector are called S-shape characteristics due to the S shape.

(B)It has the best linearity among all the FM demodulators.

(C)This circuit is difficult to tune as the three tuned circuits are to be tuned at three different frequencies.

(D)Amplitude limiting is provided.

Q.7 An FM signal is produced by a carrier signal $c(t) = 18 \cos (12\pi \times 10^6 t)$ and message signal $m(t) = \cos 4\pi \times 10^3 t$. If the peak frequency deviation of the FM signal is twice the transmission bandwidth of an AM signal which was generated by m(t) and c(t), then the coefficient of the term $\cos (12036\pi \times 10^3 t)$ in the FM signal (in terms of Bessel coefficients) is-

(A)$J_2(3)$

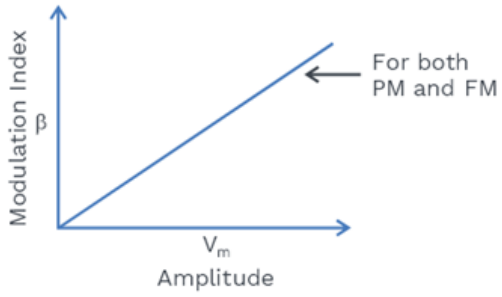
(B)$J_4(9)$

(C)$J_3(2)$

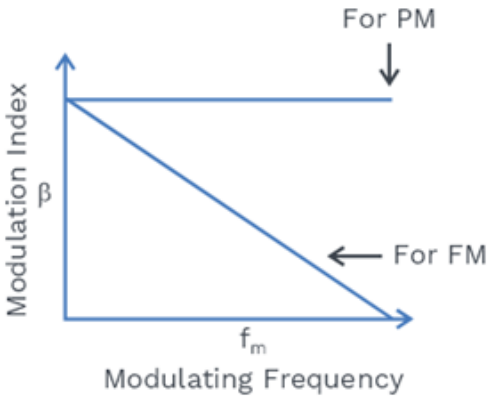
(D)$J_9(4)$

Q.8 Which of the following graph(s) illustrates the INCORRECT relationship for PM and FM modulation?

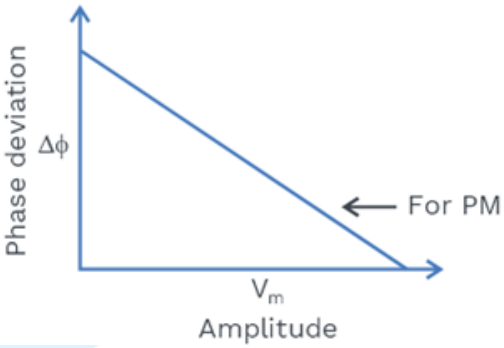
(A)



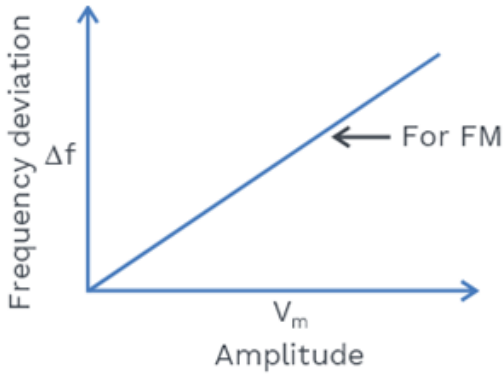
(B)



(C)



(D)



Q.9 An FM modulator has the following parameters:

Peak frequency deviation = 80 kHz

Peak amplitude of carrier signal = 10 V

Carrier frequency = 100 MHz

Modulating signal frequency = 10 kHz

The bandwidth of the FM signal is _____kHz.(Rounded off to the nearest integer).

Q.10 A PM modulator has 3.5 rad/V deviation sensitivity and the modulating signal 2.8 cos (5000 πt). The peak phase deviation is _____ rad. (up to one decimal place)

Q.11 A 20 MHz carrier is modulated by a 400 Hz audio sine wave. The maximum carrier voltage(amplitude) is 4 V and the maximum frequency deviation is 10 kHz. The phase modulated (PM) signal is expressed as-

(A)$25\sin\left(125.66\times10^{6}t+25\sin 513.27t\right)$

(B)$4\sin\left(125.66\times10^{6}t+25\sin 2513.27t\right)$

(C)$25\sin\left(125.66\times10^{6}t+4\sin 2513.27t\right)$

(D)$4\sin\left(125.66\times10^{6}t-4\sin 2513.27t\right)$

Q.12 If a carrier signal of 2400 kHz is modulated with 500 Hz and 4 kHz audio sine waves simultaneously. The frequency(in kHz) that will be absent in the output is-

(A)2404

(B)2396

(C)2399.5

(D)1200

Q.13 Which of the given statement(s) is/are correct?

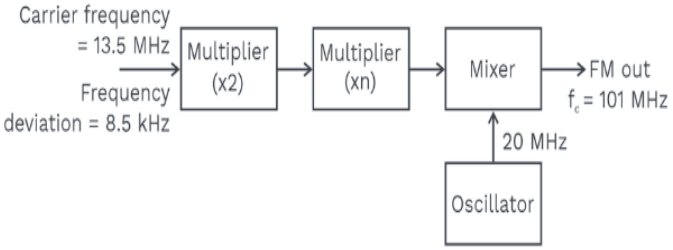
(A)In frequency modulation, the instantaneous frequency deviation is proportional to the amplitude of the modulating signal.

(B)In phase modulation, the instantaneous phase deviation is proportional to the amplitude of the modulating signal.

(C)To get FM using PM, we first integrate the baseband signal and then apply it to the phase modulator.

(D)To get PM using FM, we first differentiate the baseband signal and then apply it to the frequency modulator.

Q.14 The block diagram below illustrates a segment of an FM transmitter consisting of the frequency multiplication and mixer stages.



The input is an FM signal with a carrier frequency of 13.5 MHz and a frequency deviation of 8.5 kHz. The desired carrier frequency of FM out is 101 MHz and the mixer is designed to select the additive frequency. The value of n and frequency deviation at FM out are respectively:

(A) 5 and 15 kHz

(B)2 and 25.5 kHz

(C)3 and 25.5 kHz

(D) 3 and 51 kHz

Q.15 An angle-modulated signal is described as $s(t)=12\cos\left(\omega_c t+6\sin 1000\pi t+10\sin 2000\pi t\right)$ with $2\pi\times10^5$ rad/sec as the carrier frequency. If Δf = 12864.78 Hz, then which one of the given options is **incorrect**?

(A)The power of the angle modulated signal is 72 W.

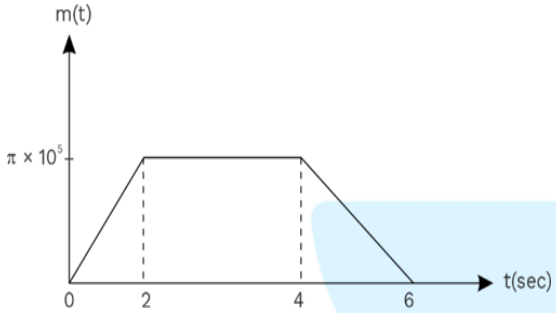
(B)The deviation ratio or modulation index is 12.864

(C)The deviation ratio or modulation index is 6.432

(D)The bandwidth of angle modulated signal is 27.729 kHz

Q.16 For a Phase modulator, the phase sensitivity, k_p = 3 rad/Volt and modulating signal m(t) = 2.4 cos (2π 2000 t). The peak phase deviation is _____rad. [Up to two decimal places]

Q.17 The message signal m(t) is shown below.



The carrier frequency is 10^6 Hz. If m(t) is phase modulated with phase deviation constant k_p = 4 rad/V, then the maximum instantaneous frequency of the modulated signal is_____ MHz.

Q.18 A modulating signal m(t) = 20sin (2π × 10^4t) is used in the phase modulation with a carrier frequency of 30 MHz. If the modulation index of frequency modulation and phase modulation are $\beta_f = \beta_p$ = 8, then for 98% power transmission-

(A)Phase deviation constant = 4 × 10^3 rad/V

(B)Phase deviation constant = 8 × 10^3 rad/V

(C)Bandwidth = 360 kHz

(D)Bandwidth = 180 kHz

Q.19 An FM wave is represented by the voltage equation

V(t) = 15sin (5×10^8t + 6sin1000t), then

(A)Frequency deviation = 592.6 Hz

(B)Frequency deviation = 954.9 Hz

(C)The power dissipated in 15Ω resistor by the FM wave is 7.5 W.

(D)The power dissipated in 25Ω resistor by the FM wave is 4.5 W.

Q.20 Which of the given statement regarding frequency modulation is true?

(A)In FM, the frequency of the carrier signal is varied with the instantaneous amplitude of the message signal.

(B)In FM, the frequency of the carrier signal is varied with the instantaneous frequency of the message signal.

(C)In FM, the amplitude of the carrier signal is varied with the instantaneous frequency of the message signal

(D)In FM, the amplitude of the carrier signal is varied with the instantaneous amplitude of the message signal.

Q.21 The power dissipated by the given FM wave in the 120Ω resistor will be –

$$S_{FM}(t) = 30 \sin\left[6 \times 10^8 t + 8 \sin 1205t\right]$$

(A)5.3 W

(B)3.75 W

(C)4.9 W

(D)2.5 W

Q.22 The bandwidth of the modulated signal when a carrier is frequency modulated with a 4 kHz sinusoidal signal and results in a 9 kHz of maximum frequency deviation will be -

(A)14 kHz

(B)26 kHz

(C)13 kHz

(D)6.5 kHz

Q.23 The angle-modulated waves are usually characterized by -

(1) Easy visualization of message waveforms

(2) Irregularity of zero-crossing.

(C) Improved noise performance.

(4) Non-linearity.

Select one of the following options-

(A)1, 2 and 3.

(B)1, 3, and 4.

(C)1 and 2 only.

(D)2, 3 and 4.

Q.24 A carrier signal of frequency 3 kHz & amplitude 10V is frequency modulated by a sinusoidal modulating signal of frequency 100 kHz and amplitude of 2 V. If the frequency deviation is 3 kHz, then the sensitivity of FM is–

(A)1.5×10^3 Hz/V

(B)3×10^3 Hz/V

(C)2.5×10^3 Hz/V

(D)2×10^3 Hz/V

Q.25 A 20-MHz carrier is a frequency modulated by a sinusoidal signal such that the maximum frequency deviation is 100 kHz. Determine the modulation index and approximate bandwidth of the FM signal, if the frequency of the modulating signal is 1 kHz.

(A)1, 100 KHz

(B)100, 200KHz

(C)10, 10KHz

(D)110, 200KHz

Q.26 Consider an angle modulated signal-

x_{PM}(t) or s(t) = 10cos (w_ct + 3sin w_mt)

Assume f_m = 1 KHz. Find the bandwidth of given modulated signal-

(A)8 KHz

(B)4 KHz

(C)12 KHz

(D)16 KHz

Q.27 A carrier signal of $6.5 \cos 4\pi \times 10^5 t$ is frequency modulated by a message signal of $4.4 \cos \left(2\pi \times 10^3 \mathrm{t}\right)$. Such that maximum frequency deviation of FM equals to 7 times of AM Bandwidth. The coefficient of the term $\cos 2\pi \times 2020 \times 10^2 t$ in resultant FM expression is _____.

(A)$6.5\ \mathrm{J}_{14}(2)$

(B)$6.5 J_2(14)$

(C)$4.4\ \mathrm{J}_2(14)$

(D)$6.5\ \mathrm{J}_2(7)$

Q.28 If a FM signal is passed through a frequency multiplier then

1. Carrier frequency is same at output

2. Message signal frequency is same at output

3. Modulation index is different at output

4. Frequency deviation changed at output

Which of the following statements are true?

(A)1 and 2 only

(B)2 and 4 only

(C)2, 3 and 4 only

(D)1, 3 and 4 only

Q.29 A sinusoidal message signal having root mean square value of 4V and frequency of 1 kHz is fed to a phase modulator with phase deviation constant 2 rad/volt. If the carrier signal is $c(t) = 2\cos(2\pi\ 10^6 t)$, the maximum instantaneous frequency of the phase modulated signal (rounded off to one decimal place) is _____ Hz.

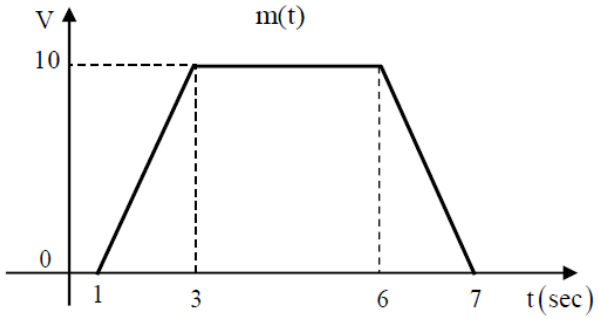
Q.30 $S_{PM}(t)$ and $S_{FM}(t)$ as defined below, are the phase modulated and the frequency modulated waveforms, respectively, corresponding to the message signal m(t) shown in the figure.

$S_{PM}(t)=\cos(1000\pi t+K_p m(t))$

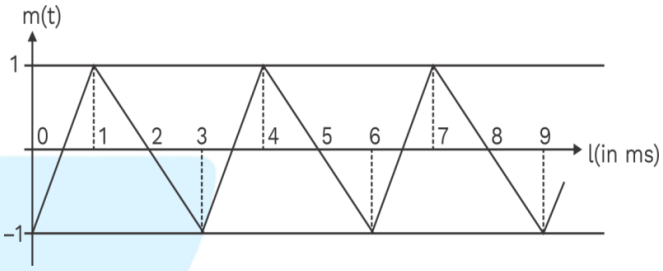
And

$S_{FM}(t)=\cos\left(1000\pi t+K_f\int_{-\infty}^{t} m(\tau)d\tau\right)$

Where K_p is the phase deviation constant in radians/volt and K_f is the frequency deviation constant in radians/second/volt. If the highest instantaneous frequencies of $S_{PM}(t)$ and $S_{FM}(t)$ are same, then the value of the ratio $\frac{K_p}{K_f}$ is _____ seconds.



Q.31 The baseband signal m(t) shown in the figure is phase-modulated to generate the PM signal $\varphi(t) = \cos(2\pi f_c t + km(t))$.



The time t on the x-axis in the figure is in milliseconds. If the carrier frequency is f_c = 50 kHz and k = 10π, the ratio of the minimum instantaneous frequency (in kHz) to the maximum instantaneous frequency (in kHz) is _____ (rounded off to 2 decimal places).

Q.32 Consider an FM signal $(t) = \cos[2\pi f_c t + \beta_1 \sin 2\pi f_1 t + \beta_2 \sin 2\pi f_2 t]$.The maximum deviation of the instantaneous frequency from the carrier frequency f_c is.

a) $\beta_1 f_1+\beta_2 f_2$

b) $\beta_1 f_2+\beta_2 f_1$

c) $\beta_1+\beta_2$

d) f_1+f_2

Q.33 An FM signal with modulation index m_f is passed through a frequency doubler. The modulation index of the output signal will be-

(A)m_f

(B)$2m_f$

(C)$4m_f$

(D)$m_f/2$

Q.34 A carrier signal of 110 MHz is frequency modulated by the sinusoidal modulating signal. 100 kHz is the maximum frequency deviation. Determine the approximate transmission bandwidth (BW) of the FM signal when the frequency of the modulating signal is 600 kHz.

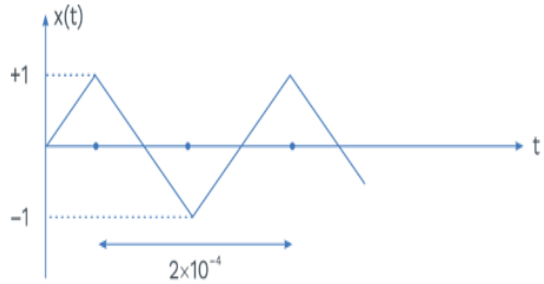
(A)120 kHz

(B)0.12 MHz

(C)21 MHz

(D)1200 kHz

Q.35 A periodic signal x(t) has bandwidth of 5 kHz and the signal is time expanded by the factor of 3 and applied to phase modulator that has phase sensitivity $K_p = 7\pi$ rad/sec. Then the bandwidth of the PM signal is _____kHz. (up to 2 decimal places).



Q.36 The minimum value of modulation index for frequency modulation required to improve the signal to noise ratio over an amplitude modulation system with modulation index of 0.5 is _____. (Up to 2 decimal places).

Q.37 In a frequency modulated signal, the value of modulation index is 0.4. If the maximum frequency of message signal is 150 kHz and the amplitude of carrier signal is 5 volts, then which of the following is/are true?

(A)Bandwidth = 300 kHz

(B)Power required for transmission = 6.5 Watts

(C)Bandwidth = 313 kHz

(D)Power required for transmission = 13.5 Watts

Q.38 An angle modulated signal is given

$$u(\mathrm{t}) = 50\cos\left[2\pi f_c t + 2\sin 2000\pi t\right]$$

Where $f_c = 48\mathrm{MHz}$. The peak – phase deviation $\Delta\phi$ is and peak – frequency deviation is Δf. Identify correct option/options.

(A)$\Delta\phi = 2\mathrm{rad}$

(B)$\Delta\phi = 4\mathrm{rad}$

(C)$\Delta f = 2000\ \mathrm{Hz}$

(D)$\Delta f = 4000\ \mathrm{Hz}$

Q.39 Which of the following statement(s) is/are correct?

(A)Bandwidth of narrow band FM is greater than bandwidth of AM signal.

(B)If the probability of any symbol is greater than others, the information that it carries is also greater than others.

(C)Noise in FM is less than noise in AM.

(D) Auto covariance function
$C_{xx}(\tau) = R_{xx}(\tau) - (\bar{X})^2$

Q.40 A carrier signal of 110 MHz is frequency modulated by the sinusoidal modulating signal. 100 kHz is the maximum frequency deviation. Determine the approximate transmission bandwidth (BW) of the FM signal

when the frequency of the modulating signal is 600 kHz.

(A)120 kHz

(B)0.12 MHz

(C)21 MHz

(D)1200 kHz

Q.41 An FM signal with frequency deviation of 80 kHz and modulating signal bandwidth of 3 kHz is applied to a device, which has input-output relation as y(t) = $x^2(t)$, where x(t) is the input and y(t) is the output. The bandwidth of the output signal is

(A)320 kHz

(B)356 kHz

(C)326 kHz

(D)346 kHz

Q.42 A message signal with amplitude and frequency 1.6 V and 10 kHz respectively modulates a carrier of frequency f_c in both AM and FM. The frequency sensitivity (k_f) is 50 kHz/volt. Find the modulating index for both AM and FM, if the amplitude is the same for both in frequency component (f_c + 10 kHz) of the spectrum.

(Given: $J_1(2) = 0.57$, $J_1(4) = -0.06$ and $J_1(8) = 0.23$)

(A)$\mu = 0.74; \beta = 4$

(B)$\mu = 0.46; \beta = 8$

(C)$\mu = 0.55; \beta = 2$

(D)$\mu = 1; \beta = 4$

Answer Key

Q.1 [6]

Q.2 [A]

Q.3 [C]

Q.4 [D]

Q.5 [A, B, C]

Q.6 [A, C]

Q.7 [D]

Q.8 [C]

Q.9 [180]

Q.10 [9.8]

Q.11 [B]

Q.12 [D]

Q.13 [A, B, C, D]

Q.14 [D]

Q.15 [C]

Q.16 [7.2]

Q.17 [1.1]

Q.18 [A, D]

Q.19 [B, C, D]

Q.20 [A]

Q.21 [B]

Q.22 [B]

Q.23 [D]

Q.24 [A]

Q.25 [B]

Q.26 [A]

Q.27 [B]

Q.28 [C]

Q.29 [1011313.7]

Q.30 [2]

Q.31 [0.75]

Q.32 [A]

Q.33 [B]

Q.34 [D]

Q.35 [46.67]

Q.36 [0.272]

Q.37 [A, D]

Q.38 [A, C]

Q.39 [C, D]

Q.40 [D]

Q.41 [C]

Q.42 [B]

3 Receiver

MCQ, MSQ and NAT

Q.1 Which of the following statement(s) is/are CORRECT for the PLL (Phase-locked loop)?

(A)It is a negative feedback system.

(B)It is used for tracking the frequency and phase of the carrier component of an incoming PM signal.

(C)It is also used in commercial FM receivers.

(D)It can demodulate FM signals even in the presence of large noise and low signal power.

Q.2 Which of the following statement(s) is/are correct for the FM capture effect?

(A)The complete suppression of the weaker signal at the receiver limiter occurs.

(B)When both signals are nearly equal in strength, then the receiver can switch from one to the other.

(C)It is the phenomenon in which only the stronger of two signals at or near the similar frequency will be demodulated.

(D)The suppression of the stronger signal occurs at the receiver limiter.

Q.3 Which one of the following is TRUE for Pre-Emphasis and De-Emphasis?

(A)The alteration of the relative strength is De-Emphasis.

(B)Pre-Emphasis is used when the signal is lost in between the transmission channel.

(C)A de-Emphasis filter is used to attenuate the low frequency boosted during pre-emphasis.

(D)Pre-Emphasis is the artificial boosting of higher modulating frequencies.

Q.4 A superheterodyne receiver has an intermediate frequency of 450 kHz, and the receiver is tuned to 700 kHz with a quality factor (Q) of 60 at the same frequency. The image rejection ratio (IRR) is approximately-

(A)75

(B)111

(C)99

(D)80

Q.5 Which of the following is not a part of the basic components in a phase-locked loop (PLL)?

(A)Multiplier

(B)Voltage-controlled oscillator

(C)Differentiator

(D)Loop filter

Q.6 A double-tuned amplifier used in a superheterodyne receiver has uncoupled tuned circuits Q = 100 each. The signal frequency is 50 MHz and the IF is 5.6 MHz. The local oscillator frequency at the mixer is greater than the incoming signal frequency.

Which of the following statement(s) is/are true?

(A)Local oscillator frequency is 50 MHz.

(B)Local oscillator frequency is 55.6 MHz.

(C)IIR is 40.71

(D)IIR is 28.18

Q.7 A superheterodyne receiver has an intermediate frequency of 550 kHz and the receiver is tuned to 900 kHz with a quality factor (Q) of 80 at the same frequency. The image rejection ratio (IRR) is_____. (Up to two decimal places).

Q.8 A broadcast superheterodyne receiver is tuned to 1500 kHz and the intermediate frequency is 450 kHz. If the image rejection ratio is 70, then the quality factor will be_____(rounded off to the nearest integer)

Q.9 A super heterodyne receiver is coupled with mixer by tuned circuit with Q factor equal to 50. If the local oscillator feeds input at 1590 kHz to the mixer and receiver is tuned to 550 kHz, then which of the following statement(s) is/are correct.

(A)Image frequency rejection ratio = 90.8

(B)Image frequency = 2630 kHz

(C)Intermediate frequency = 1040 kHz

(D)Image frequency rejection ratio = 228.50

Q.10 When a broadcast station at 1400 kHz frequency is heard simultaneously with another station of 3500 kHz frequency, without filtering, then the intermediate frequency (IF) would have been

(A)550 kHz

(B)905 kHz

(C)1000 kHz

(D)1050 kHz

Q.11 A receiver is tuned to 1260 kHz station, corresponding local oscillator frequency is 1360 kHz, and Q factor of 50. The value of image rejection ratio is _____.

Q.12 In a superheterodyne receiver, the tuned signal frequency is greater than local oscillator frequency which is 2100 MHz, and the intermediate frequency is 70 MHz. The image frequency is –

(A)1400 MHz

(B)2750 MHz

(C)1550 MHz

(D)2030 MHz

Q.13 Which of the following statement is/are true.

1. The capture range of a PLL is always less than or equal to the lock range

2. The feedback loop of the PLL has a crystal oscillator.

3. An important part of PLL is the VCO.

(A)1 and 2

(B)1 and 3

(C)2 and 3

(D)All 1, 2, 3

Q.14 In a superheterodyne receiver, which one of the following is NOT the advantage of including the RF amplifiers-

(A)Better selectivity.

(B)Good image-frequency rejection.

(C)Low power consumption.

(D)Better sensitivity.

Q.15 Consider an AM broadcast-band super heterodyne receiver having IF, RF and local oscillator frequencies as 400 kHz , 600 kHz and 1000 kHz ,respectively. The IFRR for a preselector, having Q of 100 would be -

(A)190.50

(B)200.50

(C)280.10

(D)213.13

Q.16 Arrange the following detectors in ascending order according to the linearity performance of the S-shaped characteristics of the detector:

1. Ratio Detector.

2. Slope Detector.

3. Foster Seeley Detector.

(A)1 < 2 < 3

(B)2 < 1 < 3

(C)3 < 2 < 1

(D)3 < 1 < 2

Q.17 For a receiver, the local oscillator frequency is 1970 kHz; it is tuned to an 806 kHz station. Find the intermediate and image frequency.

(A)2225 and 555 kHz

(B)1750 and 4328 kHz

(C)1164 and 3134 kHz

(D)2711 and 3734 kHz

Q.18 Amplitude limiter is used in FM modulation because -

(A)It increases the signal power.

(B)It increases the noise power.

(C)It increases the amplitude fluctuation in a certain limit.

(D)It helps in reducing noise.

Q.19 **Assertion(A):** Envelope detector works properly if modulation index is less

than 1 .

Reason(R): Envelope modulator work properly only for over-modulated signal.

(A)Both A and R are true, and R is the correct explanation of A.

(B)Both A and R are true, but R is not correct explanation of A.

(C)A is true but R is false.

(D)A is false but R is true.

Q.20 A receiver is tuned to 1350 kHz station corresponding local oscillator frequency of 1250 kHz. The image frequency of the receiver is _____.

(A)1450 kHz

(B)1550 kHz

(C)1150 kHz

(D)1050 kHz

Q.21 Consider a superheterodyne receiver tuned to 600 kHz. If the local oscillator feeds a 1000 kHz signal to the mixer, the image frequency (in integer) is _____ kHz.

Answer Key

Q.1 [A, C, D]

Q.2 [A, B, C]

Q.3 [D]

Q.4 [B]

Q.5 [C]

Q.6 [B, C]

Q.7 [141.76]

Q.8 [72]

Q.9 [B, C, D]

Q.10 [D]

Q.11 [14.48]

Q.12 [D]

Q.13 [B]

Q.14 [C]

Q.15 [A]

Q.16 [B]

Q.17 [C]

Q.18 [D]

Q.19 [C]

Q.20 [C]

Q.21 [1400]

4 Pulse Modulation Schemes

MCQ, MSQ and NAT

Q.1 An analog voltage in the range of 0-16 V is divided into 32 equal intervals to convert it to a 4-bit digital output. The maximum quantization error is -

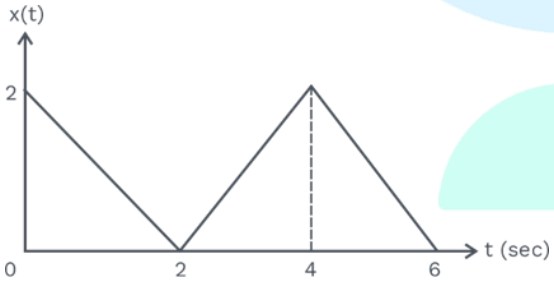
(A)0.25

(B)0.5

(C)1

(D)2

Q.2 A signal x(t) is given below.



The signal x(t) is applied as an input to the matched filter. The maximum value of the output signal is -

(A)16

(B)8

(C)4

(D)2

Q.3 A signal is uniformly distributed between [–2, 2]. The signal has undergone uniform PCM with 128 levels. The SQNR of the signal is-

(A)48.16 dB

(B)42.14 dB

(C)36.12 dB

(D)33.11 dB

Q.4 Consider an audio signal with spectral components that are limited to frequency bands 400 Hz and 4800 Hz. A PCM signal is generated with a sampling rate of 9200 samples/sec. The SQNR is 45dB. The minimum number of bits is-

(A)8

(B)7

(C)6

(D)None of the above

Q.5 An analog signal has a bandwidth of 10 kHz. The given signal is sampled at 1.5 times the Nyquist rate. The samples are quantized into 512 equally likely levels. If the successful samples are independent, then which of the following statement(s) is/are true?

(A)The information rate of the source is 30 kbps.

(B)The information rate of the source is 270 kbps.

(C)The minimum SNR required for error-free transmission over an AWGN channel having BW of 30 kHz is 27.08 dB

(D)The minimum SNR required for error-free transmission over an AWGN channel having BW of 30 kHz is 8.12 dB

Q.6 Six message signals having individual bandwidth of 3 kHz are time-division multiplied using binary PCM. The maximum error in sampling amplitude must not be greater than 1 % of peak signal amplitude. The sampling rate is 40 % above the Nyquist rate. If raised cosine pulse with a roll-off factor of 0.2 is used, then the minimum bandwidth is _____ kHz. (Up to two decimal places)

Q.7 A sampled signal varies from –4 V to 4 V. The signal is quantized using 'n' bits. The value of 'n' which ensures an RMS quantization error of less than 10 mV is _____. (Rounded off to the nearest valid integer)

Q.8 A delta modulator is designed to operate on speech signals limited to 3.4 kHz. The sampling rate is 10 times the Nyquist rate and the step size is 170 mV. The modulator is tested with a sinusoidal signal of 1.4 kHz. The maximum amplitude of the test signal to avoid slope overload is _____ V.

(Up to 3 decimal places.)

Q.9 The autocorrelation function of a stationary random process is given by –

$$R_x(\tau) = \frac{K^2}{2} e^{-|\tau|} \cos(2\pi f_0 \tau)$$

The magnitude of the random process is not greater than 8. The number of quantization levels required for an SQNR of 80 dB is _____ (Rounded off to the nearest valid integer)

[Assume K = 8].

Q.10 A signal m(t) = 2 cos 6500 πt + 8 cos 6000 πt + 6 cos 7000 πt is represented by its samples. The minimum sampling rate of the signal considering band pass spectra is_____ kHz. (Rounded off to the nearest integer)

Q.11 A sinusoidal signal is uniformly quantized using the PCM System. If the maximum value of the analog sample is represented with 0.4 % accuracy, then the minimum number of binary digits(bits) required is_____.(Rounded off to the nearest valid integer)

Q.12 A signal is sampled at 2.5 times of Nyquist rate and each sample is quantized into one of 2048 equally likely samples. Assume that the signal is bandlimited to 30 kHz. If the information rate of the source is A kbps and the minimum bandwidth required for the channel is D kHz, then $\frac{2A}{D}$ is_____ (Rounded off up to two decimal places)

[Given, signal to noise ratio is 20 dB]

Q.13 A message signal m(t)=50 sin 30t volt is given to the delta modulator having a pulse rate of 800 pulses per second. Find the step size (in volts) for perfect reconstruction.

(A)1.515 V

(B)3 V

(C)6.6 V

(D)1.875 V

Q.14 The Nyquist rate or Nyquist frequency and the Nyquist interval for $x(t) = \frac{1}{2\pi}\cos(2500\pi t) \cdot \cos(1500\pi t)$ is -

(A)Nyquist frequency = 4000 Hz

(B)Nyquist interval = 200 μsec

(C)Nyquist frequency = 5000 Hz

(D)Nyquist interval = 250 μsec

Q.15 A communication channel of bandwidth 200 kHz is required to transmit binary data of 0.25 Mbps. The bandwidth required by the communication channel, if the roll-off factor is doubled, is _____kHz. (Rounded off to the nearest integer).

Q.16 Determine the Nyquist rate for a continuous-time signal

$x(t) = 6\cos 50\pi t - 20\sin 300\pi t + 10\cos 500\pi t$

(A)50

(B)850

(C)300

(D)500

Q.17 12 voice signals bandlimited to 5.5 kHz are sampled at 8 kHz and pulse code modulated using 64 quantization levels. If signals are multiplexed by time division multiplexing using a 4-bit synchronizing word after every alternate frame, then the minimum channel bandwidth required for the transmission is_____kHz. (Rounded off to the nearest integer).

Q.18 4 pulse amplitude message signals bandlimited to 2 kHz, 15 kHz, 8 kHz, and 12 kHz are multiplexed using TDM. Calculate the minimum transmission bandwidth.

(A)120 kHz

(B)60 kHz

(C)35 kHz

(D)37 kHz

Q.19 For a PCM system, the discrete samples of an analog signal are to be quantized. If the maximum value of the analog sample is to be represented with 0.2 % accuracy, then the minimum number of binary digits(bits) required is-

(A)6

(B)7

(C)8

(D)9

Q.20 An analog signal of frequency 5kHz is sampled at a rate of 50% above the Nyquist rate. The sampled signal is quantized into 512 levels. In order to transmit the given analog signal, the minimum bandwidth required is-

(A)150 kHz

(B)270 kHz

(C)135 kHz

(D)300 kHz

Q.21 Consider 15 sinusoidal message signals. Each of them is bandlimited to 17 kHz. They are multiplexed using TDM scheme. The sampling rate of which is 150% higher than the Nyquist rate.

Now, also let the maximum quantization error be 2% of the peak amplitude of the considered message signal. The synchronization that is required per frame is equal to 6 extra bits. Find out the bit rate of the transmitter.

(A)7560 kbps

(B)7400 kbps

(C)8160 kbps

(D)8500 kbps

Q.22 A binary PAM communication system using rectangular pulses of duration T_b and amplitude of ±A to transmit digital

information at the rate of 900 kbps. The value of amplitude A required to achieve a probability of error as 10^{-16} if the one-sided PSD of AWGN is given as 10^{-6} W/Hz and $Q(9) \approx 10^{-16}$ will be -

(A)5V

(B)8.75 V

(C)6 V

(D)9 V

Q.23 A linear delta modulator has sinusoidal input modulating signal whose frequency varies from 200 Hz to 2 kHz and the input is sampled four times the Nyquist rate. The peak amplitude of the sinusoidal signal is 2V. Evaluate the minimum value of the step size in order to prevent slope overloading when the input signal frequency is 600 Hz.

(A) $\frac{3\pi}{10}$

(B) $\frac{6\pi}{10}$

(C) $\frac{3\pi}{20}$

(D) $\frac{3\pi}{2}$

Q.24 In a delta modulator based baseband system, the maximum channel bandwidth available for transmission of the binary signal is 15 kHz. The maximum frequency of the signal that can be allowed is

(A)7.5 kHz

(B)15 kHz

(C)30 kHz

(D)None of the above

Q.25 A message signal m(t) = 20sin50t V is given to the data modulator having pulse rate of 1500 pulses per second, find the step size (in mV) of output for perfect reconstruction.

Q.26 In a binary PCM system, the maximum tolerable error in sample amplitudes is 0.20% of the peak input signal amplitude. If the input signal bandwidth is 20MHz, then the maximum rate of transmission of the PCM coded bit is

(A)90 Mbps

(B)180 Mbps

(C)720 Mbps

(D)360 Mbps

Q.27 An M-ary PSK modulation scheme is used to transmit independent binary digits over a band pass channel with bandwidth 100 kHz. The bit rate is 400 Kbps, and the system characteristic is a raised cosine spectrum with 50% excess bandwidth. The minimum value of M is _____.

Q.28 In a PCM system, if input to the quantizer varies from 3V to 8V and the output of the quantizer is 6V, then the quantization error varies from

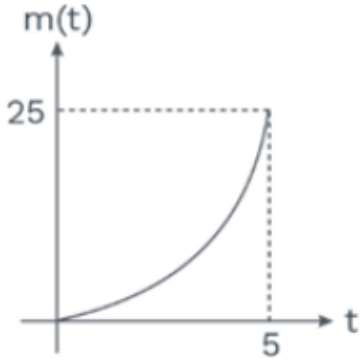
(A)–3V to 2V

(B)–2V to 3V

(C)–3V to 1V

(D)–1V to 4V

Q.29 A delta modulator has step size of 2mV. If the message signal to be modulated by using that modulator is $m(t) = t^2$, $0 \le t \le 5$, then which of the given frequencies will not result in slope overload distortion?



(A)4 kHz

(B)6.5 kHz

(C)5.5 kHz

(D)3.5 kHz

Q.30 An analog signal has a bandwidth of 4 kHz. The signal is sampled at 2.5 times the Nyquist rate and each sample is quantized into one of 256 equally likely levels. Assume that the successive samples are statistically independent. The information rate of this source (in kbps) is A _____.

For the information rate obtained, the minimum bandwidth requirement of an analog channel (in kHz) for transmitting the output of the source without errors is S/N ratio is 10 dB is B_____.

(A)A=60

(B)B=46.25

(C)A=160

(D)B=42.5

Q.31 A binary PAM communication system using rectangular pulses of duration T_b and amplitude of ±A to transmit digital information at the rate of 900 kbps. The value of amplitude A required to achieve a probability of error as 10^{-16} if the one-sided PSD of AWGN is given as $10^{-6}\ \mathrm{W/Hz}$ and $Q(9) \approx 10^{-16}$ will be -

(A)5V

(B)8.75 V

(C) 6 V

(D)9 V

Q.32 A linear delta modulator has sinusoidal input modulating signal whose frequency varies from 200 Hz to 2kHz and the input is sampled four times the Nyquist rate. The peak amplitude of the sinusoidal signal is 2V. Evaluate the minimum value of the step size in order to prevent slope overloading when the input signal frequency is 600 Hz.

(A) $\frac{3\pi}{10}$

(B) $\frac{6\pi}{10}$

(C) $\frac{3\pi}{20}$

(D) $\frac{3\pi}{2}$

Q.33 Let x(t) be a bandpass signal having frequency components of 400 Hz and 2 kHz. The sampling rate of the signal in kHz is _____.

Q.34 Find the step size required for a delta modulator operating at sampling rate of twice the Nyquist rate to track the input signal which is a sine wave having peak amplitude of 12 V and input frequency of 2 kHz. The maximum input signal frequency is 8 kHz and assume there is no slope overload.

(A) $\frac{3\pi}{2}$

(B) $\frac{\pi}{20}$

(C) $\frac{5\pi}{2}$

(D) $\frac{\pi}{40}$

Q.35 A message signal $4\cos\left(2\pi \times 10^3 t\right)$ is given to PCM system with sampling rate 25% more than Nyquist rate. Maximum quantization error should be atmost 0.2% of peak amplitude of message signal. The value of bit rate is R_b _____(in kbps).

Q.36 Consider a case of uniform quantization with quantizer output coded with 8-bit sequence. Let x(t) be a signal undergoing quantization.

If x(t) is modelled as the sample function of a zero mean Gaussian random process and is fed to a input of a quantizer with dynamic range $\pm 2\sigma_x$, where σ_x^2 is the variance of x(t) the signal to quantization noise ratio is _____ (dB)

Q.37 A linear delta modulator is designed to operate on speech signals limited to 3.4 kHz. The sampling rate is 10 times the Nyquist rate of the speech signal. The step size is 100 mV. The modulator is tested with a 1 kHz sinusoidal signal. The maximum amplitude of this test signal required to avoid slope overload is

(A)2.04 V

(B)1.08 V

(C)4.08 V

(D)2.16 V

Q.38 Find the maximum quantization error of PCM when the input signal amplitude has a range of 0 to 10 V and there are 5 bits per sample.

(A) 0.3125

(B)0.15625

(C)0.078

(D)0.625

Q.39 If a modulated signal has a modulating frequency of 16 kHz and the signal is sampled at twice the Nyquist rate and each sample is quantized to 128 equally likely levels (all samples are statistically independent). The bit rate of the source is -

(A)448 kbps

(B)424 kbps

(C)441 kbps

(D)484 kbps

Q.40 On tripling the sampling frequency-

(A)Quantization noise power decreases by 4.77 dB

(B)Quantization noise power increases by 4.77 dB

(C)Quantization noise power is unchanged

(D)Quantization noise power can be increased or decreased by 4.77 dB

Q.41 What will be the time duration (in nsec) of 1 bit of binary encoded signal, if an analog signal is sampled at 5 kHz and the number of quantizing levels is 128?

(A)14285

(B)7143

(C)21428

(D)28571

Q.42 In a binary pulse code modulation system, the maximum tolerable error in sample amplitudes is 0.25% of the peak input signal amplitude. If the input signal bandwidth

is 10MHz, then the minimum rate of transmission of the PCM coded bits is-

(A)60 Mbps

(B)180 Mbps

(C)150 Mbps

(D)90 Mbps

Q.43 The bit rate and band rate of an analog signal are 4000 bps and 500 baud respectively. The data elements carried by each signal and signal elements needed are respectively-

(A) 4 and 16

(B)8 and 256

(C)2 and 4

(D)7 and 128

Answer Key

Q.1 [A]

Q.2 [B]

Q.3 [B]

Q.4 [A]

Q.5 [B, C]

Q.6 [211.68]

Q.7 [9]

Q.8 [1.314]

Q.9 [13]

Q.10 [1]

Q.11 [8]

Q.12 [13.32]

Q.13 [D]

Q.14 [A, D]

Q.15 [275]

Q.16 [D]

Q.17 [296]

Q.18 [D]

Q.19 [C]

Q.20 [C]

Q.21 [C]

Q.22 [C]

Q.23 [C]

Q.24 [B]

Q.25 [666.67]

Q.26 [D]

Q.27 [64]

Q.28 [A]

Q.29 [B, C]

Q.30 [B, C]

Q.31 [C]

Q.32 [C]

Q.33 [4]

Q.34 [A]

Q.35 [22.5]

Q.36 [46.92]

Q.37 [A]

Q.38 [B]

Q.39 [A]

Q.40 [A]

Q.41 [D]

Q.42 [B]

Q.43 [B]

5 Digital Shift Keying Techniques

MCQ, MSQ and NAT

Q.1 A 50 MHz carrier is QPSK modulated. The carrier is being modulated by a 1.674 Mbps T_1 data stream. A raised cosine filter with $\alpha = 0.2$ is employed by the transmitter. The transmission bandwidth of the signal is -

(A)2008.8 kbps

(B)1004.4 kbps

(C)502.2 kbps

(D)251.1 kbps

Q.2 To have a bit rate of 16 kbps in a coherent binary FSK system, the minimum bandwidth required is -

(A)128 kbps

(B)64 kbps

(C)32 kbps

(D)16 kbps

Q.3 Binary PSK is employed to transmit information over an AWGN channel having the power spectral density of $N_0 = 2 \times 10^{-10}$ W/Hz. The energy of transmitting signal is $E_b = A^2\frac{T}{2}$; where T = bit interval and A = amplitude of the signal(in Volts)

Consider x = 4.75. When the data rate is 20 kbps, then to achieve an error probability of 10^{-6}, the amplitude of the signal is _____ $\times 10^{-3}$. (Up to 1 decimal place)

(Assume $Q(x) = 10^{-6}$ for x = 4.75)

Q.4 Consider a 16-PSK system operating with are information bit rate of 48kbps. Which of the following statement is/are true?

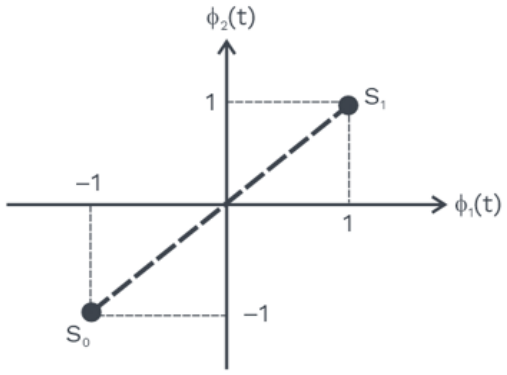
(A)The baud rate of the system is 12×10^3 symbols/sec

(B)The baud rate of the system is 3×10^3 symbols/sec

(C)Bandwidth efficiency is 12 bits per sec per cycle.

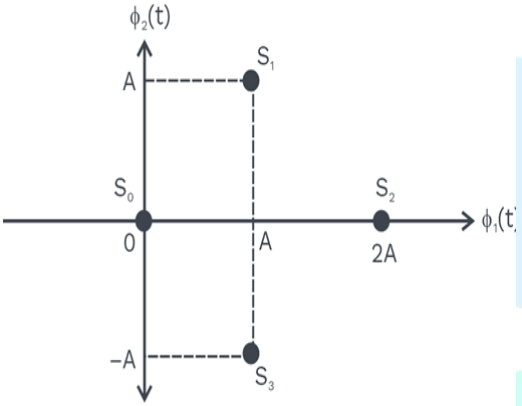
(D)Bandwidth efficiency is 4 bits per sec per cycle.

Q.5 The constellation diagram of a binary modulation scheme is shown below. The two equiprobable symbols are transmitted through an AWGN channel of two-sided noise $\text{PSD} = \frac{1}{2}\ \text{W/Hz}$. If a correlator receiver with optimum threshold detection is used at the receiver end, then the bit error rate (BER) of the system will be Q(x) where x = _____. (Rounded off to the nearest integer)



$$P(S_0) = P(S_1) = \frac{1}{2}$$

Q.6 Consider the signal space representation of 4-ary modulation scheme as shown below.



If the symbols S_0, S_1, S_2 & S_3 are occurring with probabilities of 0.4, 0.2, 0.1, and 0.3 respectively, then the average symbol energy of the modulation scheme is equal to _____ (Up to 3 decimal places)

[Assume A = 0.7]

Q.7 A transponder is using QPSK modulation with a bandwidth of 72 MHz and the roll-off factor of raised cosine pulses is 0.8. The maximum data rate(in Mbps) is_____ .(Up to 1 decimal place)

Q.8 A message signal in DPSK system is given as m(n) = {1, 1, 0, 1, 1, 0 ……..}. Find the phase of the carrier.

[Consider the initial value of the storage element as 1].

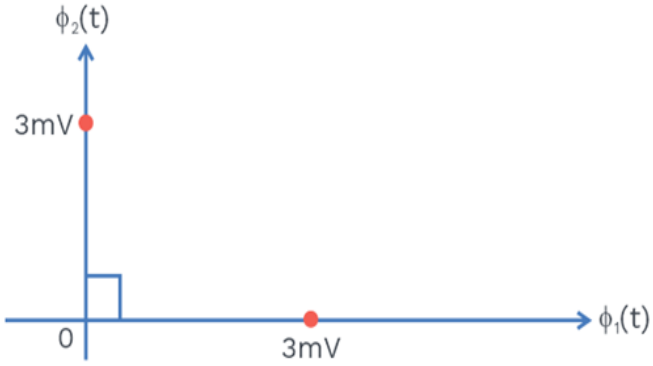
(A)$\{\pi, 0, \pi, \pi, 0, \pi \cdots\}$

(B)$\{\pi, 0, 0, \pi, 0, 0 \ldots\}$

(C)$\{0, 0, \pi, 0, 0, \pi \ldots\}$

(D)$\{\pi, \pi, 0, \pi, \pi, 0 \ldots\}$

Q.9 The constellation diagram of a BFSK modulation scheme is given below.



The modulated signal is transmitted through an AWGN channel with a two-sided PSD of 4 × 10^{-8} W/Hz. The signal is received by the correlator receiver with exact phase synchronization. If both messages are equiprobable, then the bit error rate (BER) of the system in terms of Q-function is–

(A)$Q[5\sqrt{2}]$

(B)$Q[10\sqrt{3}]$

(C)$Q[7.5\sqrt{2}]$

(D)$Q[5.5\sqrt{3}]$

Q.10 The two coherent modulation schemes BPSK and BFSK have the same bit error rates (BERs). If the energy per bit of the systems is E_{BPSK} and E_{BFSK} respectively, then the ratio of energies $\frac{5E_{BPSK}}{3E_{BFSK}}$ is_____ (Up to two decimal places)

Q.11 The minimum required bandwidth for a non-coherently detected orthogonal binary FSK system with a higher frequency signaling tone of 1 MHz and the symbol duration of 1 ms will be-

(A)1 kHz

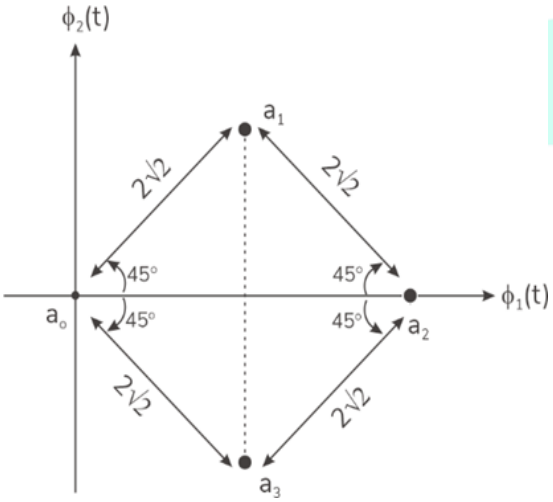
(B)3 kHz

(C)1 MHz

(D)3 MHz

Q.12 The average bit energy used in a coherent BPSK system is E_{BPSK} and the average bit energy used in a coherent BFSK system is E_{BFSK}. If these two systems are having same bit error rate (BER) under identical circumstances, then the ratio of energies $\frac{E_{BFSK}}{E_{BPSK}}$ should be equal to_____. (Rounded off to the nearest integer)

Q.13 Consider the 4-ary modulation scheme as shown in the figure below.



The probabilities of a_o, a_1, a_2, and a_3 are 0.3, 0.3, 0.2, and 0.2. Then the average symbol energy(in Joules) of the modulation scheme will be –

(A)8.4

(B)6.6

(C)12

(D)7.2

Q.14 A transponder with bandwidth of 56 MHz and 0.5 roll off factor, find the maximum data rate that can be transmitted using QPSK modulation.

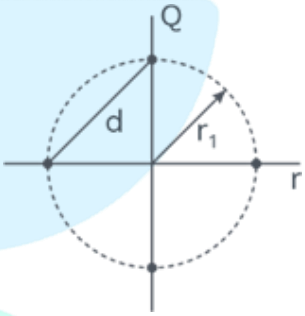
(A)74.6 bps

(B)74.6 kbps

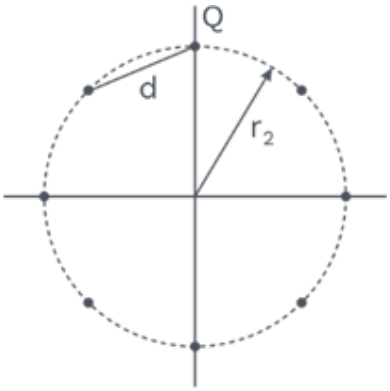
(C)74.6 Mbps

(D)74.6 Gbps

Q.15 Two signal constellations are as shown below.



Four-phase signal constellation



Eight phase signal constellation

If $r_1 = 0.707$ d and $r_2 = 1.307$d, d being the minimum distance between pair of signal

points, then assuming the SNR to be high, the additional average transmitted signal energy required by the 8-PSK signal to achieve the same error probability as the 4-PSK signal is _____ dB.

(up to 2 decimal places). Assume that 'd' is same is both the constellations and all the signals are equally probable.

Q.16 In coherent schemes, a small pilot is added for synchronization because the pilot does not carry information, it causes degradation in probability of bit error. Consider a coherent PSK that uses the following two pulses of duration T_b

$$p(t) = A\sqrt{1-m^2}\cos\omega_c t + Am\sin\omega_c t$$

$$q(t) = -A\sqrt{1-m^2}\cos\omega_c t + Am\sin\omega_c t$$

→ Where $\mathbf{Am\sin\omega_c t}$ is the pilot (Assuming channel noise to be white Gaussian) probability of Error will be

→ Where A= 5V

$$T_b = 1\mu sec$$

$$\frac{No}{2} = 2.5\times10^{-6}\ W/Hz$$

$$m = \frac{1}{2}$$

(A)$Q[\sqrt{5}]$

(B)$Q[\sqrt{10}]$

(C)$Q[\sqrt{3.75}]$

(D)$Q[\sqrt{2}]$

Q.17 A binary PSK transmits $s_1(t) = A\cos\omega_c t; 0 \le t \le T$ for bit '1' and $s_2(t) = A\cos(\omega_c t + \pi); 0 \le t \le T$ for bit '0'. At correlator receiver, the local carrier wave from is $A\cos(\omega_c t + \phi)$ instead of $A\cos\omega_c t$ due to poor synchronization. Let $(A^2\ T/\eta) = 10, \phi = 60^\circ$ and T = n/f; n is an integer. Then which of the following conclusions can be drawn.

(A)Probability of Bit Error is $Q[\sqrt{2.5}]$

(B)Probability of Bit Error is Q[2.5]

(C)Probability of Bit Error is $Q[\sqrt{10}]$

(D)Bandwidth Efficiency is 0.5

Q.18 A BPSK scheme operating over an AWGN channel with noise power spectral density of $\frac{N_0}{2}$, uses equiprobable signals $s_1(t) = \sqrt{\frac{2E}{T}}\sin(\omega_c t)$ and $s_2(t) = -\sqrt{\frac{2E}{T}}\sin(\omega_c t)$ over the symbol interval (0, T). If the local oscillator in a coherent receiver is ahead in phase by 45° with respect to the received signal, then the probability of error in the resulting system is

(A) $Q\left(\sqrt{\frac{2E}{N_0}}\right)$

(B) $Q\left(\sqrt{\frac{E}{N_0}}\right)$

(C) $Q\left(\sqrt{\frac{d_{min}^2}{2N_0}}\right)$, where d_{min} is the minimum distance between 2 symbols.

(D) $Q\left(\sqrt{\frac{d_{min}^2}{4N_0}}\right)$, where d_{min} is the minimum distance between 2 symbols.

Q.19 Find the bandwidth of a BFSK system with two frequencies 35 kHz and 40 kHz for a bit duration of 0.25 mS.

(A)12 kHz

(B)11 kHz

(C)13 kHz

(D)14 kHz

Q.20 The bandwidth of the channel using BPSK technique is 200 kHz, find the bandwidth that will be required for 16-PSK transmission.

(A)100 kHz

(B)200 kHz

(C)50 kHz

(D)400 kHz

Q.21 A QPSK modulated channel requires bandwidth-

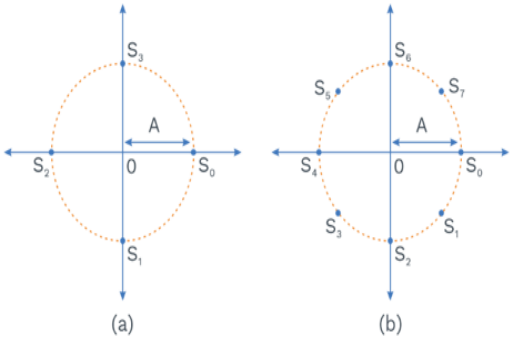
(A) Equal to BPSK

(B)Equal to FSK

(C)Half of the bandwidth of BPSK

(D)Double the bandwidth of BPSK

Q.22 Consider the statement about given constellation diagrams (if all symbols are equiprobable)



1. P_e (constellation 1) $> P_e$ (constellation 2)
2. P_e (constellation 2) $> P_e$ (constellation 1)
3. Average symbol energy is different in both constellation
4. Average symbol energy is same in both constellation

(A)1 and 3 only

(B)1 and 4 only

(C)2 and 3 only

(D)2 and 4 only

Q.23 A symbol stream contains alternate QPSK and 16-QAM symbols. If symbols from this stream are transmitted at the rate of 1 mega-symbols per second, the raw (uncoded) data rate is _____mega-bits per second (rounded off to one decimal place).

Answer Key

Q.1 [B]

Q.2 [B]

Q.3 [9.5]

Q.4 [A, D]

Q.5 [2]

Q.6 [0.686]

Q.7 [80]

Q.8 [B]

Q.9 [C]

Q.10 [0.83]

Q.11 [B]

Q.12 [2]

Q.13 [D]

Q.14 [C]

Q.15 [5.34]

Q.16 [C]

Q.17 [A, D]

Q.18 [B, D]

Q.19 [C]

Q.20 [C]

Q.21 [C]

Q.22 [D]

Q.23 [3]

6 Random Variables and Noise

MCQ, MSQ and NAT

Q.1 Consider the probability density function $f_X(x) = pe^{-2q|x|}$, where X is a random variable whose allowed values range from $-\infty \le X \le \infty$.

The probability density function that lies between x=2 and x=5 is-

(A) $\left[\frac{e^{-2q} - e^{-5q}}{2}\right]$

(B) $2\left(e^{-10q} - e^{-4q}\right)$

(C) $\left[\frac{e^{-4q} - e^{-10q}}{2}\right]$

(D) $2\left(e^{-5q} - e^{-2q}\right)$

Q.2 The probability mass function of a random variable is zero except at the points x = 0, 1, 2, 3. It has the value P(0) = $a^3 + 2a$, P(1) = $2a - 5a^2$,and P(2) = $2a^3 - 5a^2 + 5a - 1$ with a > 0. The value of 'a' will be

(A)3

(B) $\frac{1}{3}$

(C)2

(D) $\frac{2}{3}$

Q.3 The CDF of a random variable X is given by-

$$F_x(x) = \begin{cases} 2x & ; \quad x < 0 \\ \frac{x^2}{12} & ; \quad 0 \le x < 3 \\ (3x^2 - 2) & ; \quad x \ge 3 \end{cases}$$

The value of $P(x > 2 \mid x < 4)$ is_____. (Up to three decimal places).

Q.4 Let X be a random variable with PDF

$$f_X(x) = \begin{cases} 3; & 0 < x < 1 \\ 0; & otherwise \end{cases}$$

If y = - $5\log_e 2x$, then the PDF of random variable Y is

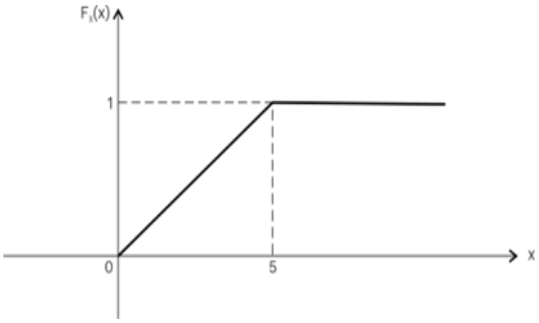
(A) $f_Y(y) = \begin{cases} \frac{1}{6}e^{-y/5} & ; \quad 0 < y < \infty \\ 0 & ; \quad otherwise \end{cases}$

(B) $f_Y(y) = \begin{cases} \frac{3}{10}e^{-y/5} & ; \quad 0 < y < \infty \\ 0 & ; \quad otherwise \end{cases}$

(C) $f_Y(y) = \begin{cases} \frac{3}{10}e^{-5y} & ; \quad 0 < y < \infty \\ 0 & ; \quad otherwise \end{cases}$

(D) $f_Y(y) = \begin{cases} \frac{1}{6}e^{-5y} & ; \quad 0 < y < \infty \\ 0 & ; \quad otherwise \end{cases}$

Q.5 The cumulative distribution function(CDF) of a random variable 'X' is given as-



Which of the following is/are correct?

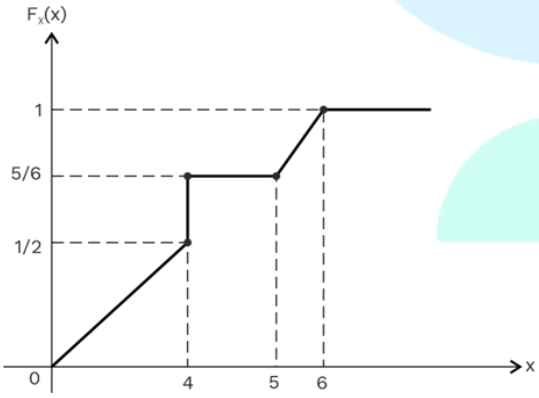
(A)P(x = 2) = 0.4

(B)P(x > 4) = 0.2

(C) $P\left(\frac{x>4}{x>2}\right)=\frac{1}{2}$

(D) $P\left(\frac{x>4}{x>2}\right)=\frac{1}{3}$

Q.6 The cumulative distribution function(CDF) of random variable 'X' is shown below.



Choose the correct statement(s) regarding probability?

(A)P(5) = 0

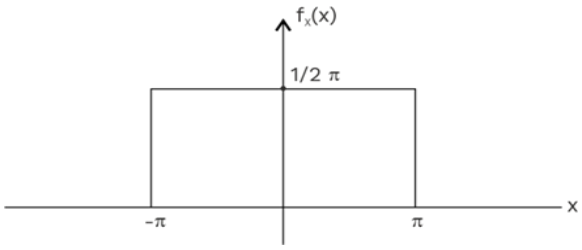
(B) $P(5)=\frac{5}{6}$

(C) $P\{4\le x<5\}=0$

(D) $P\{4\le x<5\}=\frac{1}{3}$

Q.7 The PDF of a uniformly distributed random variable 'X' is given below.

If $P=E\left[\cos\frac{x}{2}\right]-E[\cos x]$, then the value of $5P\pi$ is_____. (Rounded off to the nearest integer).



Q.8 A zero mean stationary random process V(t) has autocorrelation function given by $R_V(\tau) = 8e^{-8}$. The mean square value of random process y(t) = 5V(t) + 7 is_____. (Rounded off to the nearest integer).

Q.9 A certain random signal has an autocorrelation function, $R_X(a)=25e^{-(6a)^3}+12$. The total power of the signal is_____W. (Rounded off to the nearest integer)

Q.10 A non-bandlimited signal x(t) having power spectral density $S_X(f)=e^{-|f/5000|},(f\ in\ Hz)$, is passed through an ideal and unity gain low pass filter with a cut-off frequency of f_x Hz. The value of f_x(in kHz) such that the filter passes at least 85% of signal power at its output, is_____. (Up to one decimal place).

Q.11 For a real random process x(t), power spectral density for positive frequencies is given by $S_{XX}(\omega)=5[u(\omega-5)-u(\omega-10)]+2\delta(\omega-2)$. The average power of the random process x(t) is_____ Watts.

Q.12 Find the mean of the given Rayleigh density function.

$$f_X(x) = \begin{cases} \frac{x}{\sigma^2} e^{\frac{-x^2}{2\sigma^2}} & ; \quad x > 0 \\ 0 & ; \quad Otherwise \end{cases}$$

(A)$\sigma\sqrt{2\pi}$

(B)$\frac{\sigma}{2}\sqrt{\pi}$

(C)$2\sigma\sqrt{\pi}$

(D)$\sigma\sqrt{\frac{\pi}{2}}$

Q.13 A WSS random process X(t) with mean μ_x passes through an LTI system with impulse response $h(t) = be^{-at}u(t)$. It is desired that the for a = 2, the mean of output is $\mu_Y = 5\mu_X$.The value of b is_____. (Rounded off to the nearest integer)

Q.14 A zero mean stationary random process x(n) is having variance 3. Another random process is defined as g(n) = 2x(n) – x(n + 7). If $R_{XX}(7) = 3$, then

(A)$\sigma^2_{g(n)} = 0.5\sigma^2_{x(n)}$

(B)$\sigma^2_{g(n)} = \sigma^2_{x(n)}$

(C)$\sigma^2_{g(n)} = 7\sigma^2_{x(n)}$

(D)$\sigma^2_{g(n)} = 11\sigma^2_{x(n)}$

Q.15 The random variable 'X' takes values '0' and '1' with probability 'a' and 'b' respectively. The variance of the random variable is-

(A)b(1 – a)

(B)ab

(C)b^2

(D)ab^2

Q.16 Two independent random variables X and Y are uniformly distributed in the range [0, 3] and [0, 6] respectively. A random variable Z is defined as Z = X + Y. The mean square value of Z is _____. (Rounded off to the nearest integer)

Q.17 The input random process to an LTI system is X(t) whose impulse response is h(t). The output random process is defined by Y(t). The white noise with two-sided PSD of $\frac{N_0}{2}$ is applied as an input to the LTI system.

Then the cross-correlation function defined as $R_{XY}(\tau) = E[X(t + \tau) Y(t)]$ is equal to -

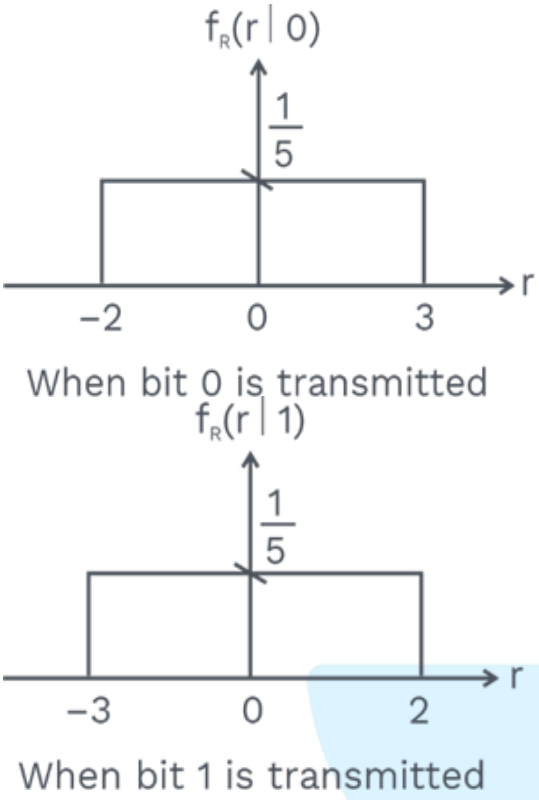
(A)$\frac{N_0}{2}h(\tau)$

(B)$N_0\delta(\tau)$

(C)$N_0h(-\tau)$

(D)$\frac{N_0}{2}\ h(-\tau)$

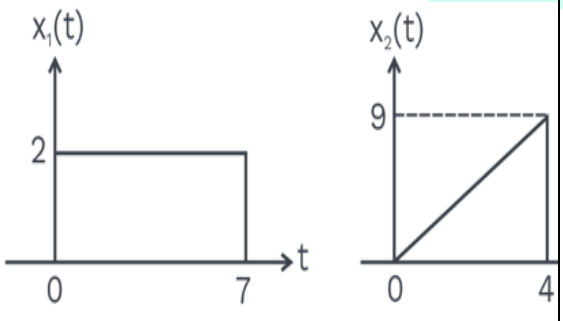
Q.18 In a digital communication system, bits '0' and '1' are transmitted through a channel with probabilities $\frac{1}{4}$ and $\frac{3}{4}$ respectively. The receiver decides the bits 1 or 0 based on the received value (R). The CDF of the received value (R) is given as-



The decision is made in favor of 0 when r < 0

The decision is made in favor of 1 when r > 0

Q.19 The correlation coefficient of the two signals shown in the figure below is_____
(Up to three decimal places)



Q.20 The autocorrelation function of a stationary random process X(t) is given as-

$$R_x(\tau) = e^{-\beta|\tau|}; \beta > 0$$

X(t) is applied to an LTI system having the impulse response $h(t) = e^{-t}u(t)$; $\alpha > 0$. The power spectral density of the output Y(t) is:

(A) $\left(\frac{\beta}{(\alpha^2 + 4\pi^2 f^2)(\beta^2 + 4\pi^2 f^2)}\right)$

(B) $\left(\frac{2\beta}{(\alpha^2 + 4\pi^2 f^2)(\beta^2 + 4\pi^2 f^2)}\right)$

(C) $\frac{\beta}{\alpha - \beta}\left[\frac{1}{(\beta + 4\pi f)^2} - \frac{1}{(\alpha + 4\pi f)^2}\right]$

(D) $\frac{2\alpha}{\beta^2 - \alpha^2}\left[\frac{1}{\alpha^2 + 4\pi^2 f^2} - \frac{1}{\beta^2 + 4\pi^2 f^2}\right]$

Q.21 A WSS random process X(t) having a mean value of 6.34 is applied to the input of an LTI system with the impulse response $h(t) = \frac{3}{2}e^{-3\alpha t}u(t)$. If the output Y(t) of the system has a mean of $\frac{9}{4}$, then the value of α is-

(A)0

(B)0.72

(C)1.41

(D)2.18

Q.22 The joint probability density of the random variables X and Y is $f(x, y) = Ne^{-(x+y)}$ ranging from $3 \leq x \leq \infty$, $2 \leq y \leq \infty$ and f (x, y) = 0 otherwise.

Which of the following statement(s) is/are true?

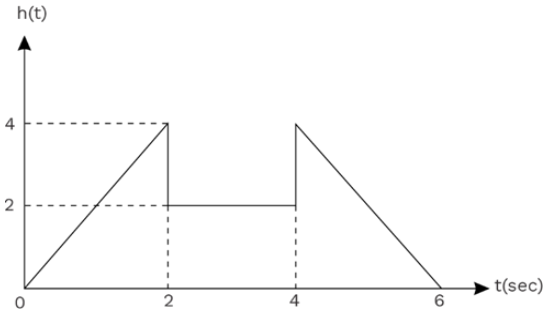
(A)The value of constant N is 6.74×10^{-3}

(B)The value of constant N is 148.41

(C)The probability P ($6 \leq x \leq 8$, $2 \leq y \leq 3$) is 0.74

(D)The probability P ($6 \leq x \leq 8$, $2 \leq y \leq 3$) is 0.0272

Q.23 A stationary random process X(t) is applied to an LTI system, whose impulse response is shown in the figure below.



The output process of this LTI system is Y(t). If mean value of the process X(t) is 4, then the mean value of Y(t) will be equal to _____. (Rounded off to the nearest integer)

Q.24 The probability of error in the transmission of digital data through a noisy channel is P_e = 0.01. If each transmission is an independent event, and for 100 transmissions only two error occurs, then the probability of error is-

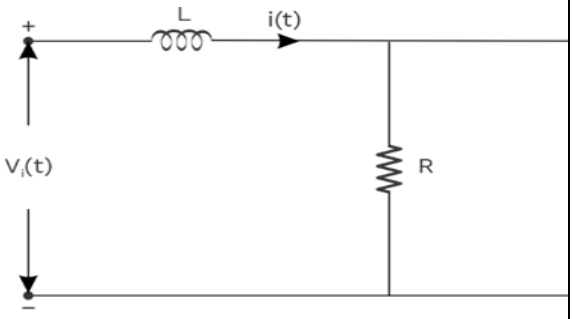
(A)0.01

(B)0.05

(C)0.08

(D)0.012

Q.25 Consider the circuit shown below.



The input has two-sided power spectral density given as $S_{V_i}(f) = \frac{N_0}{2} W/Hz$.

The average power in $V_0(t)$ is-

(A) $\frac{N_0 R}{2}$ Watts

(B) $\frac{N_0 R}{4L}$ Watts

(C) $\frac{N_0 2L}{R}$ Watts

(D) $\frac{N_0 RL}{4}$ Watts

Q.26 A noisy signal X(t) = $\cos(\omega_0 t + \theta)$ + N(t) is passed through a linear circuit with frequency response $H(\omega) = \frac{R/L}{j\omega + R/L}$, Where N(t) = zero mean noise process with auto-correlation function $R_n(\tau) = k\delta(\tau)$, ω_0 = known constant and θ = random variable uniformly distributed over $(\pi, 2\pi)$. If N(t) and θ are statistically independent, then the mean value of X(t) is-

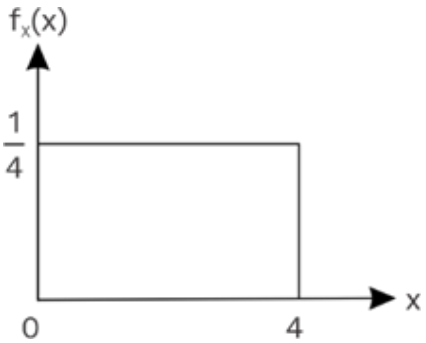
(A)0

(B)1

(C) $\frac{2}{\pi}\cos\omega_0 t$

(D) $\frac{2}{\pi}\sin\omega_0 t$

Q.27 The probability density function of a random variable 'X' is shown below.



The differential entropy h(X) of random variable 'X' is-

(A)2 bits/symbol

(B)1 bit/symbol

(C)0 bits/symbol

(D)None of the above

Q.28 The input and output random variables to a communication channel are X and Y respectively. The joint probability mass function is-

Y \ X	-1	0	1
-1	0.25	0.125	0
0	0	0.25	0
1	0	0.125	0.25

The value of $P\left(\frac{X=1}{Y=1}\right)$ is_____. (Up to two decimal places).

Q.29 A random process X(t) is defined by X(t) = $2\cos(2\pi t + Y)$

Where, Y = discrete random variable

Which of the following statement(s) is/are TRUE?

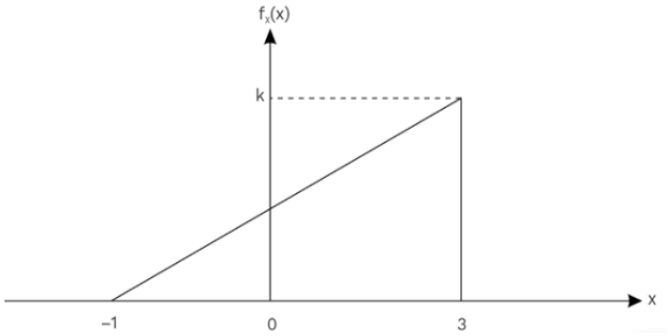
(A)$\mu_x = 1$

(B)$\mu_x = \frac{1}{2}$

(C)$R_{XX}(t_1, t_2)$ for $t_1 = 0$ and $t_2 = 1$ is 2

(D)$R_{XX}(t_1, t_2)$ for $t_1 = 0$ and $t_2 = 1$ is 1

Q.30 The mean square value of the random variable X, whose PDF is shown below is-



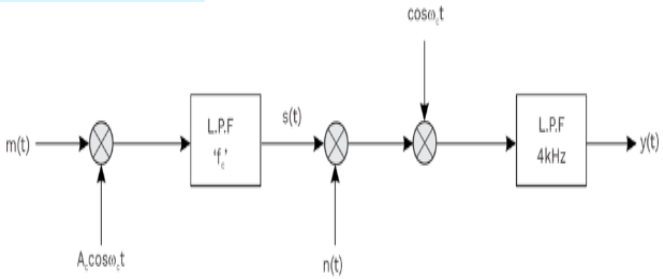
(A)5/4

(B)11/3

(C)13/4

(D)19/3

Q.31 An SSB system is shown in the figure below



Where s(t) = the lower sideband modulated signal

The Fourier transform of m(t), i.e. M(f) is given as-

$$M(f) = \begin{cases} 0.004 & ; \quad |f| \leq 2\text{kHz} \\ 0.002 & ; \quad 2\text{kHz} \leq \text{f} \leq 4\text{kHz} \\ 0 & ; \quad |f| > 4\text{kHz} \end{cases}$$

(A)The power of the signal s(t) is 200 mW when the value of A_C is 10 V.

(B)The power of the signal s(t) is 200 mW when the value of A_C is 25 V

(C)If n(t) is AWGN with PSD of $S_N(f) = \frac{N_0}{2} = 10^{-4}\frac{\text{mW}}{\text{Hz}}$, then noise power in the demodulator is 0.1 mW.

(D)If n(t) is AWGN with PSD of $S_N(f) = \frac{N_0}{2} = 10^{-4}\frac{\text{mW}}{\text{Hz}}$, then noise power in the demodulator is 0.2 mW.

Q.32 A matched filter has a frequency response given by $H(f) = \frac{1 - e^{-j10\pi f}}{j6\pi f}$. Find the impulse response h(t) of the filter.

(A) $h(t) = \frac{1}{2}[\text{sgn}(t) - \text{sgn}(t-3)]$

(B) $h(t) = \frac{1}{6}[\text{sgn}(t) - \text{sgn}(t-5)]$

(C) $h(t) = \frac{1}{5}[\text{sgn}(2t) - \text{sgn}(t-2)]$

(D) $h(t) = \frac{1}{3}[\text{sgn}(5t) - \text{sgn}(t-3)]$

Q.33 Two resistors of 200 Ω and 300 Ω are connected in series at temperatures of 300K and 400K respectively. The equivalent noise temperature (in K) is-

(A)300

(B)400

(C)450

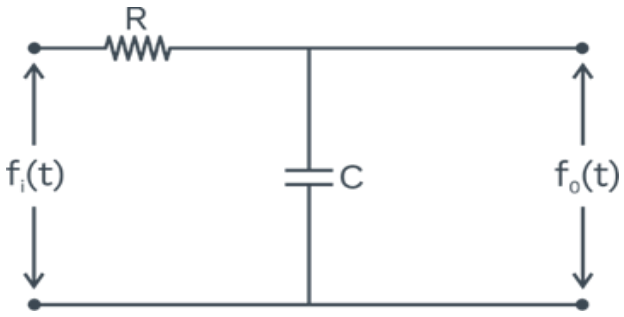
(D)360

Q.34 Consider the signal given below.

$$s(t) = \begin{cases} 3e^{-2t}; & 0 < t < 2.5\text{sec} \\ 0; & otherwise \end{cases}$$

If this signal is applied to the matched filter, then the peak value of the filter output signal is_____. (Up to two decimal places)

Q.35 An RC low pass filter is shown below.



The input $f_i(t)$ is an AWGN signal with power spectral density (double-sided) as B W/Hz. The mean square value of the output $f_0(t)$ is -

(A) $\frac{B}{4RC}$

(B) $\frac{B}{2RC}$

(C) $\frac{B}{RC}$

(D) $\frac{4B}{RC}$

Q.36 A white noise with two-sided power spectral density of $9N_0'$ is passed through an ideal differentiator. The differentiator output is passed through an ideal low pass filter having f_c = B Hz. The noise power(in Watts) at the LPF output is-

(A) $5\pi^2 N_0' B^3$

(B) $12\pi^2 N_0' B^3$

(C) $24\pi^2 N_0' B^3$

(D) $25\pi^2\ \text{N}_0'\text{B}^3$

Q.37 A noise is passed through an ideal BPF with a 2 MHz of bandwidth & centered at 60 MHz . The noise process has PSD given as -

$$S_n(f) = \begin{cases} 10^{-8}\left(1 - \frac{|f|}{10^3}\right), & |f| \le 10^8 \\ 0, & |f| > 10^8 \end{cases}$$

What will be the power content of the output process?

(A)16 mW

(B)2 W

(C)8 W

(D)6 mW

Q.38 If two random process X(t) & Y(t) are defined as-

$X(t) = A + Bt$ at $t \in [0, \infty]$

$Y(t) = A + Ct$ at $t \in [0, \infty]$

Where A, B and C are independent Normal N(1, 1) random variables.

The cross-correlation function $R_{XY}(t_1, t_2)$ and cross-co-variance function $C_{XY}(t_1, t_2)$ for t_1, $t_2 \in [0, \infty)$ at t = 100 sec are respectively.

(A)$2 + t_{-1} + t_2 + t_1t_2$ and 0

(B)$2 - t_1 - t_2 - t_1t_2$ and 1

(C)$2 + t_1 + t_2 + t_1t_2$ and $(1 + t_1)(1 + t_2)$

(D)$2 + t_1 + t_2 + t_1t_2$ and 1

Q.39 An AWGN (voice graded) telephone channel has a bandwidth of 5 kHz. The two-sided noise power spectral density is $\frac{\eta}{2} = 3.5 \times 10^{-5}$ W/Hz. Determine the minimum bit-energy required if information is to be transmitted at 60 kbps bit rate over this channel with small bit error rate.

(A)23.88 mJ/bit

(B)20.81 J/bit

(C)20.87 mJ/bit

(D)23.87 J/bit

Q.40 X and Y are independent random variables. X follows a binomial distribution, with N = 5 and $P = \frac{1}{2}$. Y takes integer values 1 and 2, with equal probability. Then the probability, that X = Y is _____. (Correct up to 2 decimal places)

Q.41 The power spectral density of a stationary random process X(t) is given by

$S_x(f) = 10e^{-2|f|} W/Hz$.

The average power of the random process X(t) is-

(A)5W

(B)10W

(C)20W

(D)25W

Q.42 The process X(t) is WSS, then $E\left[[X(t+\tau) - X(t)]^2\right]$ where $R_{xx}(\tau)$ is the auto correlation of X(t) is -

(A)$2\left[R_{xx}(0) + R_{xx}(\tau)\right]$

(B)0

(C)$2\left[R_{xx}(0) - R_{xx}(\tau)\right]$

(D)$R_{xx}(0) - R_{xx}(\tau)$

Q.43 A Gaussian channel has a bandwidth of 2 kHz and noise power spectral density of 10^{-14} watts/Hz. The signal power at the receiver is required to be not less than 0.1 mW. The channel capacity (in kbps) is approximately–

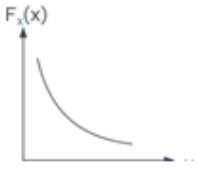
(A)51

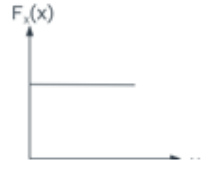
(B)49

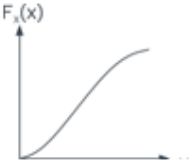
(C)81

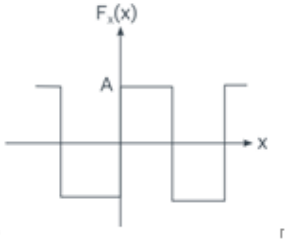
(D)20

Q.44 Which of the following is a plot of cumulative distribution function (CDF) i.e., $F_x(x)$.

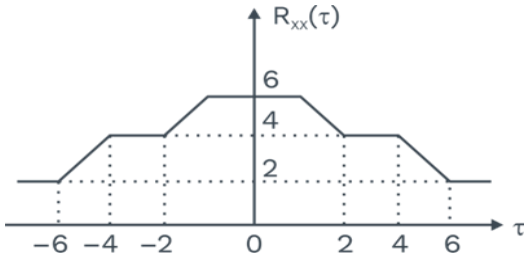
(A) 

(B) 

(C) 

(D) 

Q.45 A random process X(t) whose ACF [$[R_{xx}(\tau)]$] is given below, X(t) being stationary and ergodic, the power of the signal is _____.



(A) 36 W

(B) 12 W

(C)6 W

(D)3W

Q.46 A WSS random process X(t) has a mean value of 2.25, is applied to the input of an LTI system with impulse response $h(t) = 2e^{-at}u(t)$. The output Y(t) of the system has a mean of $\frac{1}{2}$ and is also a WSS process. The value of a is _____.

(A)4.5

(B)9

(C)225

(D)2

Q.47 X(t) is a stationary process that is passed through an LTI system with impulse response $h(t) = 5\delta(t-3)$. The auto-correction function of X(t) is $R_x(\tau) = e^{-|\tau|}$, find the output auto-correction function.

(A)$R_Y(\tau) = 5e^{-|\tau|}$

(B)$R_Y(\tau) = 15e^{-|\tau|}$

(C)$R_Y(\tau) = 25e^{-|\tau|}$

(D)$R_Y(\tau) = 25e^{-|\tau-3|}$

Q.48 A random variable X is uniformly distributed from –1.5 to 3.5. The mean square value and variance are respectively

(A)1, 2

(B)3, 2

(C)3, 4

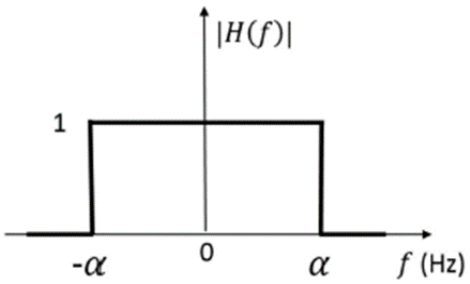
(D)3.5, 2.5

Q.49 The frequency response $H(f)$ of a linear time-invariant system has magnitude as shown in the figure.

Statement I: The system is necessarily a pure delay system for inputs which are bandlimited to $-\alpha \le f \le \alpha$.

Statement II: For any wide-sense stationary input process with power spectral density $S_X(f)$, the output power spectral density $S_Y(f)$ obeys $S_Y(f) = S_X(f)$ for $-\alpha \le f \le \alpha$.

Which one of the following combinations is true?



(A)Statement I is correct, Statement II is correct

(B)Statement I is correct, Statement II is incorrect

(C)Statement I is incorrect, Statement II is correct

(D)Statement I is incorrect, Statement II is incorrect

Q.50 Two continuous random variables X and Y are related as-

Y = 2X + 3

Let σ_X^2 and σ_Y^2 denote the variances of X and Y, respectively. The variance are related as -

(A) $\sigma_Y^2 = 2\sigma_X^2$

(B) $\sigma_Y^2 = 4\,\sigma_X^2$

(C) $\sigma_Y^2 = 5\,\sigma_X^2$

(D) $\sigma_Y^2 = 25\,\sigma_X^2$

Q.51 A binary random variable X takes the value +2 or –2. The probability P(X=+2)=α. The value of α (rounded off to one decimal place), for which the entropy of X is maximum, is _____.

Q.52 The random variable

$$Y = \int_{-\infty}^{\infty} W(t)\phi(t)dt, \quad \text{where}$$

$$\phi(t) = \begin{cases} 1, & 5 \le t \le 7 \\ 0, & \text{otherwise} \end{cases}$$

and W(t) is a real white Gaussian noise process with two-sided power spectral density $S_W(f)$ = 3 W/Hz for all f. The variance of Y is _____.

Q.53 A single bit, equally likely to be 0 and 1, is to be sent across an additive white Gaussian noise (AWGN) channel with power spectral density $N_0/2$.Binary signaling, with $0 \rightarrow p(t)$ and $1 \rightarrow q(t)$,is used for the transmission, along with an optimal receiver that minimizes the bit-error probability.

Let $\varphi_1(t)$, $\varphi_2(t)$ form an orthonormal signal set.

If we choose $p(t) = \varphi_1(t)$ and $q(t) = -\varphi_1(t)$, we would obtain a certain bit-error probability P_b. If we keep $p(t) = \varphi_1(t)$, but take $q(t) = \sqrt{E}\,\varphi_2(t)$,for what value of E would we obtain the same bit-error probability P_b?

(A) 3

(B) 1

(C) 2

(D) 0

Q.54 A random variable X takes values –1 and +1 with probabilities 0.2 and 0.8, respectively. It is transmitted across a

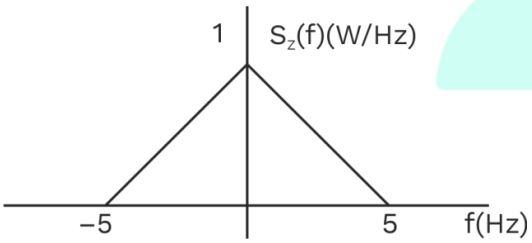
channel which adds noise N, so that the random variable at the channel output is Y = X +N. The noise N is independent of X and is uniformly distributed over the interval [–2, 2]. The receiver makes a decision

$$\hat{X} = \begin{cases} -1, \text{if } Y \le \theta \\ +1, \text{if } Y > \theta \end{cases}$$

Where the threshold $\theta \in [-1, 1]$ is chosen so as to minimize the probability of error $Pr[X \neq X]$.

The minimum probability of error, rounded off to 1 decimal place, is _____.

Q.55 Let a random process Y(t) be described as Y(t) * X(t) + Z(t), where X(t) is a white noise process with power spectral density $S_X(f)$ = 5W/Hz. The filter h(t) has a magnitude response given by |H(f)| = 0.5 for $-5 \le f \le 5$, and zero elsewhere. Z(t) is a stationary random process, uncorrelated with X(t), with power spectral density as shown in the figure.



The power in Y(t), in watts, is equal to _____ W. (rounded off to two decimal places).

Q.56 Let x_1, x_2 and x_3 be independent and identically distributed random variables with the uniform distribution on [0, 1]. The probability $P(x_1 + x_2 \le x_3)$ is _____.

Q.57 Let (t) be a wide sense stationary (WSS) random process with power spectral density $S_X(f)$. If (t) is the process defined as $Y(t) = X(2t - 1)$, the power spectral density $S_Y(f)$ is.

a) $S_Y(f)=\frac{1}{2}S_X\left(\frac{f}{2}\right)e^{-j\pi f}$

b) $S_Y(f)=\frac{1}{2}S_X\left(\frac{f}{2}\right)e^{-\frac{j\pi f}{2}}$

c) $S_Y(f)=\frac{1}{2}S_X\left(\frac{f}{2}\right)$

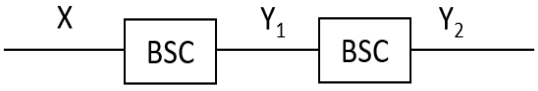
d) $S_Y(f)=\frac{1}{2}S_X\left(\frac{f}{2}\right)e^{-j2\pi f}$

Q.58 A real band-limited random process (t) has two-sided power spectral density

$$S_X(f)=\begin{cases} 10^{-6}(3000-|f|) \text{ Watts /Hz} & \text{for } |f| \le 3\text{kHz} \\ 0 & \text{otherwise} \end{cases}$$

where f is the frequency expressed in Hz. The signal (t) modulates a carrier $\cos 16000\pi t$ and the resultant signal is passed through an ideal band-pass filter of unity gain with centre frequency of 8 kHz and band-width of 2 kHz. The output power (in Watts) is _____.

Q.59 A binary random variable X take the value of 1 with probability 1/3. X is input to a cascade of 2 independent identical binary symmetric channels (BSCs) each with crossover probability ½. The output of BSCs are the random variable Y_1 and Y_2 as shown in the figure.



The value of $H(Y_1)+H(Y_2)$in bits is ___ _____.

Answer Key

Q.1 [C]

Q.2 [B]

Q.3 [0.992]

Q.4 [B]

Q.5 [B, D]

Q.6 [A, D]

Q.7 [10]

Q.8 [249]

Q.9 [37]

Q.10 [9.5]

Q.11 [8.6]

Q.12 [D]

Q.13 [10]

Q.14 [B]

Q.15 [B]

Q.16 [24]

Q.17 [D]

Q.18 [0.6]

Q.19 [0.654]

Q.20 [B]

Q.21 [C]

Q.22 [B, D]

Q.23 [48]

Q.24 [C]

Q.25 [B]

Q.26 [D]

Q.27 [A]

Q.28 [0.67]

Q.29 [A, C]

Q.30 [B]

Q.31 [A, D]

Q.32 [B]

Q.33 [D]

Q.34 [2.25]

Q.35 [B]

Q.36 [C]

Q.37 [A]

Q.38 [D]

Q.39 [A]

Q.40 [0.468]

Q.41 [B]

Q.42 [C]

Q.43 [B]

Q.44 [C]

Q.45 [C]

Q.46 [B]

Q.47 [C]

Q.48 [B]

Q.49 [C]

Q.50 [B]

Q.51 [0.5]

Q.52 [6]

Q.53 [A]

Q.54 [0.1]

Q.55 [17.5]

Q.56 [0.167]

Q.57 [C]

Q.58 [2.5]

Q.59 [2]

7 Information Theory and Coding

MCQ, MSQ and NAT

Q.1 For a (6, 3) linear block code, the parity check matrix is given below.

$$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

For a data word [1 0 1], the valid codeword is-

(A)C = [1 0 1 1 0 0]

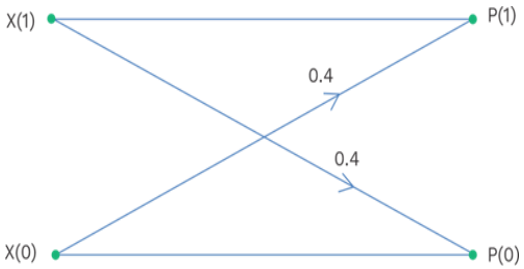
(B)C = [1 0 1 0 1 1]

(C)C = [1 1 0 1 1 0]

(D)C = [1 1 0 0 1 1]

Q.2 A binary symmetric channel(BSC) with corresponding probabilities is shown in the figure below. The probability of transmitting 1 is 0.6 and transmitting 0 is 0.4.

If P(1) is the probability of receiving 1 and P(0) is the probability of receiving 0 at the receiver, then



(A)P(1) = 0.52

(B)P(0) = 0.48

(C)P(1) = 0.58

(D)P(0) = 0.32

Q.3 A communication channel has a joint probability matrix P(X, Y) as shown below.

$$P(X,Y) = \begin{bmatrix} 0.10 & 0.15 & 0 \\ 0 & 0.10 & 0.15 \\ 0.15 & 0 & 0.35 \end{bmatrix}$$

The mutual information I(X ; Y) is-

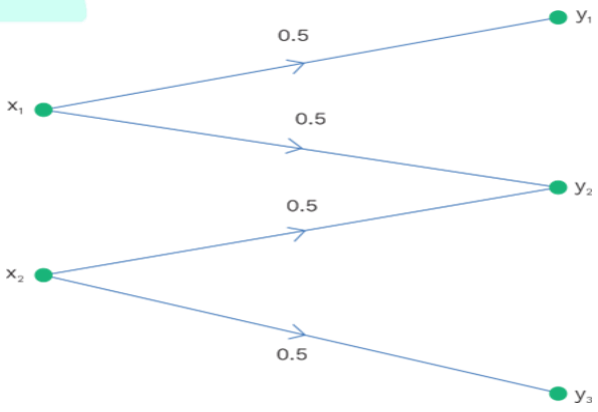
(A)1.5 bits

(B)0.5745 bits

(C)2.4259 bits

(D)0.1259 bits

Q.4 The channel diagram shown below has two equiprobable inputs x_1 and x_2, and outputs are y_1, y_2, and y_3. The output entropy H(Y)[in bits per symbol] is-



(A)3.5

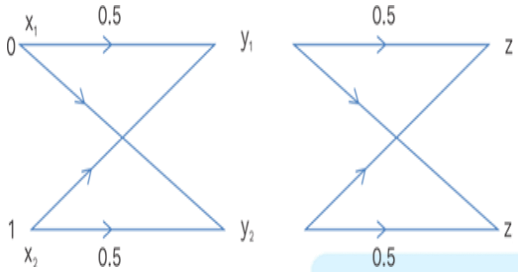
(B)2.5

(C)1.5

(D)1.25

Q.5 A binary memoryless channel has inputs x_0 and x_1 and outputs y_0 and y_1. $P\left(\frac{y_0}{x_1}\right) = 0.9, P\left(\frac{y_1}{x_0}\right) = 0.1$ and $P(x_0) = 0.5$. The probability $P\left(\frac{x_0}{y_0}\right)$ is _____. (Up to two decimal places).

Q.6 A cascaded system of two binary symmetric channels is shown below.



If P(0) = 0.3 and P(1) = 0.7, then $3P(y_2) + 5P(z_1)$ is _____. (Rounded off to the nearest integer).

Q.7 BCD codes are used to transmit data into a channel. The code efficiency for BCD code is_____. (Up to two decimal places).

Q.8 X is an exponential random variable. The PDF is defined as-

$$f_x(x) = \begin{cases} \frac{1}{r} e^{-\frac{x}{r}} & ; \quad x > 0 \\ 0 & ; \quad \text{otherwise} \end{cases}$$

Where r > 0

The differential entropy of the continuous random variable X is-

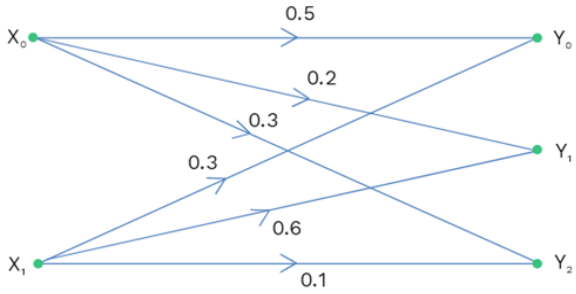
(A)$r\log_2(r)$

(B)$1.44 + \log_2(r)$

(C) $\ln(r) + \frac{1}{\ln(2)}$

(D)None of the above

Q.9 Consider the system shown in the figure below.



X_1 and X_0 are equiprobable. The value of $P\left(\frac{X_1}{Y_1}\right)$ is_____. (Up to two decimal places).

Q.10 A (5, 2) linear block code is defined by the given codewords.

E = {00000, 01001, 10111, 01111}

The minimum distance of the code (d_{min}) is_____. (Rounded off to the nearest integer)

Q.11 A (7, 4) linear block code is having parity matrix, $\bar{P}$ as-

$$\bar{P} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

The received codeword [1 1 0 1 1 0 1] is having a single-bit error. The transmitted codeword is-

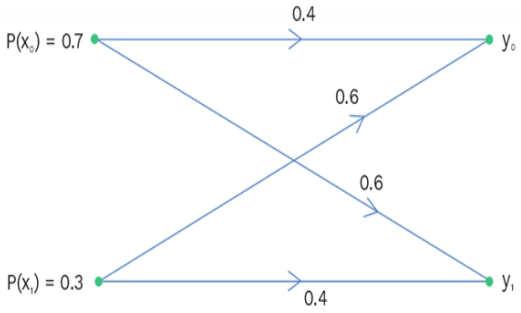
(A)[0 1 0 1 1 0 1]

(B)[1 1 1 1 1 0 1]

(C)[1 0 0 1 1 0 1]

(D)[1 1 0 0 1 0 1]

Q.12 Consider the Binary symmetric channel(BSC) shown below.



The decision device connected to the receiver side is optimized using MAP criteria. The probability of error in the decision-making is-

(A)$P(x_0)$

(B)$P(x_1)$

(C)0.03

(D)0.072

Q.13 A discrete memoryless source transmits 5 symbols with probabilities 0.5, 0.25, 0.125, 0.0625, and 0.0625. The symbols are coded into a binary format using 3 different codewords. The coding schemes are shown below.

x_i	$P(x_i)$	Coding(A)	Coding(B)	Coding(C)
x_0	0.5	1	00	111
x_1	0.25	10	01	11
x_2	0.125	110	10	10
x_3	0.0625	1110	11	01
x_4	0.0625	1111	110	00

Which of the following statement(s) is/are true?

(A)The average codeword length for A is less than that of B.

(B)The average codeword length for C is 2.5 bits/symbol.

(C)The most efficient coding scheme is A.

(D)the average codeword length for C is less than that of B.

Q.14 A discrete memoryless source produces 5 symbols with probabilities $\frac{1}{2}, \frac{1}{5}, \frac{3}{20}, \frac{1}{10}$ and $\frac{1}{20}$. The symbols are coded using Huffman coding.

Which of the following statement(s) is/are true?

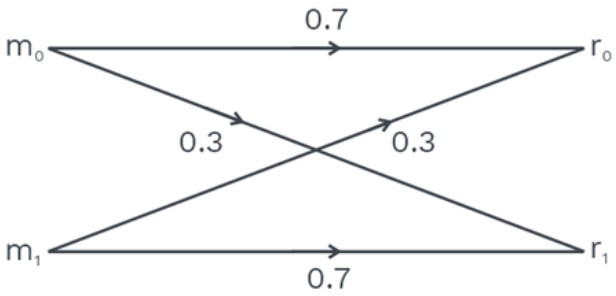
(A)The average length of the codeword is 1.95 bits/symbol

(B)Entropy is 1.78 bits/symbol

(C)Redundancy is 0.98

(D)Redundancy is 0.014

Q.15 A binary symmetric channel(BSC) is shown in the figure below.



The symbols m_0 and m_1 are transmitted by the source with probabilities of 0.3 and 0.7 respectively. The probability $P(m_1 \mid r_0)$ is equal to_____. (Up to two decimal places)

Q.16 A discrete memoryless source emits 5 symbols 'xi' with probabilities P(xi) where i = 0 to 4. A source coding scheme converts symbols into the binary format as shown below.

x_i	$P(x_i)$	Code
x_0	0.1	00
x_1	0.1	01
x_2	0.1	10
X_3	0.3	100
x_4	0.4	1000

Which of the following statement(s) is/are true?

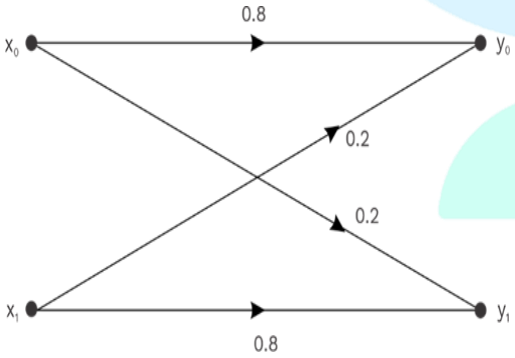
(A)Entropy = 2.046 bits/symbol

(B)Average codeword length = 3.1 bits /symbol

(C)Efficiency = 96%

(D)Efficiency = 66%

Q.17 Consider the binary symmetric channel as shown in the figure below.



If $P(x_0) = 0.6$ and $P(x_1) = 0.4$, then the probability to find the transmitted symbol x_1 when y_0 is received is_____. (Up to three decimal places).

Q.18 A memoryless source emits 3000 binary symbols/sec and each symbol has a probability of 0.3 when transmitted 0 and 0.7 when transmitted 1.

The minimum number of bits required for error-free transmission is-

(A)987.16 bits/sec

(B)1221.93 bits/sec

(C)2078.12 bits/sec

(D)2643.87 bits/sec

Q.19 A memoryless source has an alphabet A = {-4, -2, -1, 0, 1, 2, 4} with corresponding probabilities given as,

$$P(A) = \left\{\frac{1}{20}, \frac{1}{10}, \frac{1}{10}, \frac{3}{20}, \frac{1}{20}, \frac{1}{4}, \frac{3}{10}\right\}.$$

Which of the following statement(s) is/are true?

(A)The entropy of source is 2.528 bits/symbol

(B)The entropy of source is 5.282 bits/symbol.

(C)The source is quantized as

$$\begin{cases} q(-4) = q(-2) = -3 \\ q(-1) = q(0) = q(1) = 0 \\ q(2) = q(4) = 3 \end{cases}$$

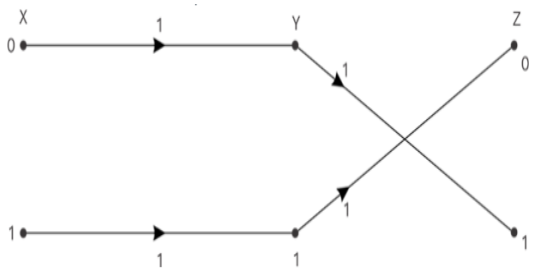
The entropy of the quantized source is 1.406 bits/symbol.

(D)The source is quantized as

$$\begin{cases} q(-4) = q(-2) = -3 \\ q(-1) = q(0) = q(1) = 0 \\ q(2) = q(4) = 3 \end{cases}$$

The entropy of the quantized source is 2.523 bits /symbol.

Q.20 A cascaded communication system is shown below.



Assuming P(0) = P(1), the probability of error for transmission in the communication system shown above is_____. (Rounded off to the nearest integer)

Q.21 A binary channel matrix is given by-

$$P\left[\frac{Y}{X}\right] = \begin{matrix} & Y_0 & Y_1 \\ X_0 & \frac{3}{4} & \frac{1}{4} \\ X_1 & \frac{3}{10} & \frac{7}{10} \end{matrix}$$

If $P(X_0) = \frac{1}{4}$ and $P(X_1) = \frac{3}{4}$, then the ratio of $\frac{P(Y_1)}{P(Y_0)}$ is_____. (Up to two decimal places)

Q.22 A channel has an SNR of 63 and a bandwidth of 15 kHz. The channel capacity is _____. (in kbps) (Up to two decimal places)

Q.23 A black and white TV picture has about 2×10^6 picture elements with 64 brightness levels. They are repeated at the rate of 8 elements per second. If all elements are independent of each other and all levels are equiprobable, the rate of information by the TV picture is _____.

(A) 96 Mbps

(B) 100 Mbps

(C)140 Mbps

(D)81 Mbps

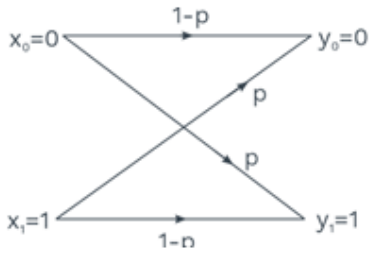
Q.24 Consider the following statements -

1. I(X; Y) =0 when input and output symbols of a channel are statistically independent.

2. Mutual information can be expressed as-

$$I(X;Y) = \sum_{j=0}^{r-1}\sum_{k=0}^{k-1} p(x_j, y_k) \log_2 \frac{p(x_j, y_k)}{p(x_j)\, p(y_k)}$$

3. Binary symmetric channel is shown as—



4. Baye's rule for conditional probabilities is-

$$\frac{p(x_1/y_k)}{p(x_j)} = \frac{p(y_k/x_j)}{p(y_k)}$$

Which of the following statements are correct?

(A) 1 and 2 only

(B) 1 , 2 and 3 only

(C) 2 , 4 and 5 only

(D) 1, 2 , 3 and 4

Q.25 Consider the given statements-

(1) The entropy of a continuous random variable 'X' with a probability density function fX(x), can be expressed as-

$$h(x) = \int_{-\infty}^{\infty} f_x(x) \log_2 \left[\frac{1}{f_x(x)}\right] dx$$

, where h(X) is the differential entropy of X.

(2) Differential entropy of a continuous random variable is always positive.

(3) For finite variance(σ^2), the Gaussian random variable has the smallest differential entropy (in comparison to any random variable).

(4) I(X; Y) = h(X) – h(X/Y) where X and Y are a pair of continuous random variables and h(X/Y) is the conditional differential entropy.

Which of the following statements is/are true-

(A)1 and 2 only

(B)2 and 4 only

(C)1 and 4 only

(D)1 , 2 and 4 only

Q.26 consider the following statements about channel matrix-

$$p\left(\frac{Y}{X}\right) = \begin{bmatrix} p_1 & p_2 \\ p_3 & p_4 \end{bmatrix}$$

(1) For p2 = p3 = 0 and p1 = p4 = 1; the channel is noiseless.

(2) For p3 + p4 = 1; p3≠ 0, p4 ≠ 0 and p1 = p2 = 0; the channel is deterministic.

(3) For p1 = p4 = 0 and p2 = p3 = 1; the channel is lossless and deterministic

(4) For p1 = p3 = 1 and p2 = p4 = 0, the channel is noiseless.

Which of the following statements is/are true?

(A) 1 and 2 only

(B) 1 and 3 only

(C) 3 and 4 only

(D) 2 , 3 and 4 only

Q.27 A source produces 3 symbols with equal probability calculate the entropy of the source.

(A)1.58

(B)0.33

(C)1.43

(D)0.477

Q.28 Let H(X) denote the entropy of a discrete random variable X taking K possible distinct real values. Which of the following statements is/are necessarily true?

(A)$H(X) \leq \log_2 K$

(B)H(X) ≤ H(2X)

(C)$H(X) \leq H\left(X^2\right)$

(D)$H(X) \leq H\left(2^X\right)$

Q.29 A digital transmission system uses a (7,4) systematic linear Hamming code for transmitting data over a noisy channel. If three of the message-codeword pairs in this code (m_i ; c_i), where c_i is the codeword corresponding to the ith message m_i, are known to be (1100; 0101100), (1110; 0011110) and (0110; 1000110), then which of the following is a valid codeword in this code?

(A) 1101001

(B) 1011010

(C) 0001011

(D) 0110100

Q.30 A digital communication system transmits a block of N bits. The probability of error in decoding a bit is α. The error event of each bit is independent of the error events of the other bits. The received block is declared erroneous if at least one of its bits is decoded wrongly. The probability that the received block is erroneous is:

(A) $N(1-\alpha)$

(B) α^N

(C) $1-\alpha^N$

(D) $1-(1-\alpha)^N$

Q.31 A linear Hamming code is used to map 4-bit messages to 7-bit codewords. The Encoder mapping is linear. If the message 0001 is mapped to the codeword 0000111, and the message0011 is mapped to the codeword 1100110, then the message 0010 is mapped to:

(A) 0010011

(B) 1111111

(C) 1111000

(D) 1100001

Answer Key

Q.1 [B]

Q.2 [A, B]

Q.3 [B]

Q.4 [C]

Q.5 [0.5]

Q.6 [4]

Q.7 [0.83]

Q.8 [B]

Q.9 [0.75]

Q.10 [2]

Q.11 [A]

Q.12 [B]

Q.13 [A, B, C]

Q.14 [A, D]

Q.15 [0.5]

Q.16 [A, B, D]

Q.17 [0.143]

Q.18 [D]

Q.19 [A, C]

Q.20 [1]

Q.21 [1.42]

Q.22 [89.99]

Q.23 [A]

Q.24 [D]

Q.25 [C]

Q.26 [B]

Q.27 [A]

Q.28 [A, B, D]

Q.29 [C]

Q.30 [D]

Q.31 [D]